



Tow-fish and Furious

A LOW-COST PROTON MAGNETOMETER TOW-FISH FOR DETECTION OF UNDERWATER WRECKAGE IN SHALLOW COASTAL WATERS

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Background Information

How the Project Came to Be:

- This project was motivated by recovery efforts off the coast of Vietnam, where shipwrecks, downed aircraft, and missing soldiers remains.
- Nonprofit organizations perusing this initiative lacked access to affordable, customizable, and scalable magnetometer systems needed for extended search missions.
- This project aims to address that gap by developing a low-cost proton magnetometer tow-fish to support more accessible and effective recovery operations.



Project Goal:

Design and implement a cost-effective proton magnetometer tow-fish system that follows these requirements:

- Maintains sufficient sensitivity to detect magnetic anomalies
- Reduces overall system cost
- Can be deployed from small vessels
- Supports humanitarian and recovery missions

Core Functionality

► First Phase: Polarization

- A strong current is passed through a coil surrounding a hydrogen-rich fluid such as water
- This aligns the spins of the protons

► Second Phase: Precession

- The current is then turned off
- Protons begin to spin around the magnetic field

► Third Phase: Signal Detection

- The precession induces a small AC voltage in the coil
- This signal produces a decaying sinusoid

► Fourth Phase: Signal Amplification & Filtering

- The induced signal is extremely weak
- It is amplified using a low-noise amplifier
- Filtering removes unwanted noise and interference
- Produces a clean and usable signal

► Fifth Phase: Frequency Measurement

- The frequency of precession is directly proportional to the magnetic field strength
- The system then extracts this frequency to compute the field

► Sixth Phase: Data Processing & Storage

- The magnetic field data is processed and saved to an SD card
- Stored data can later be used for mapping and analysis

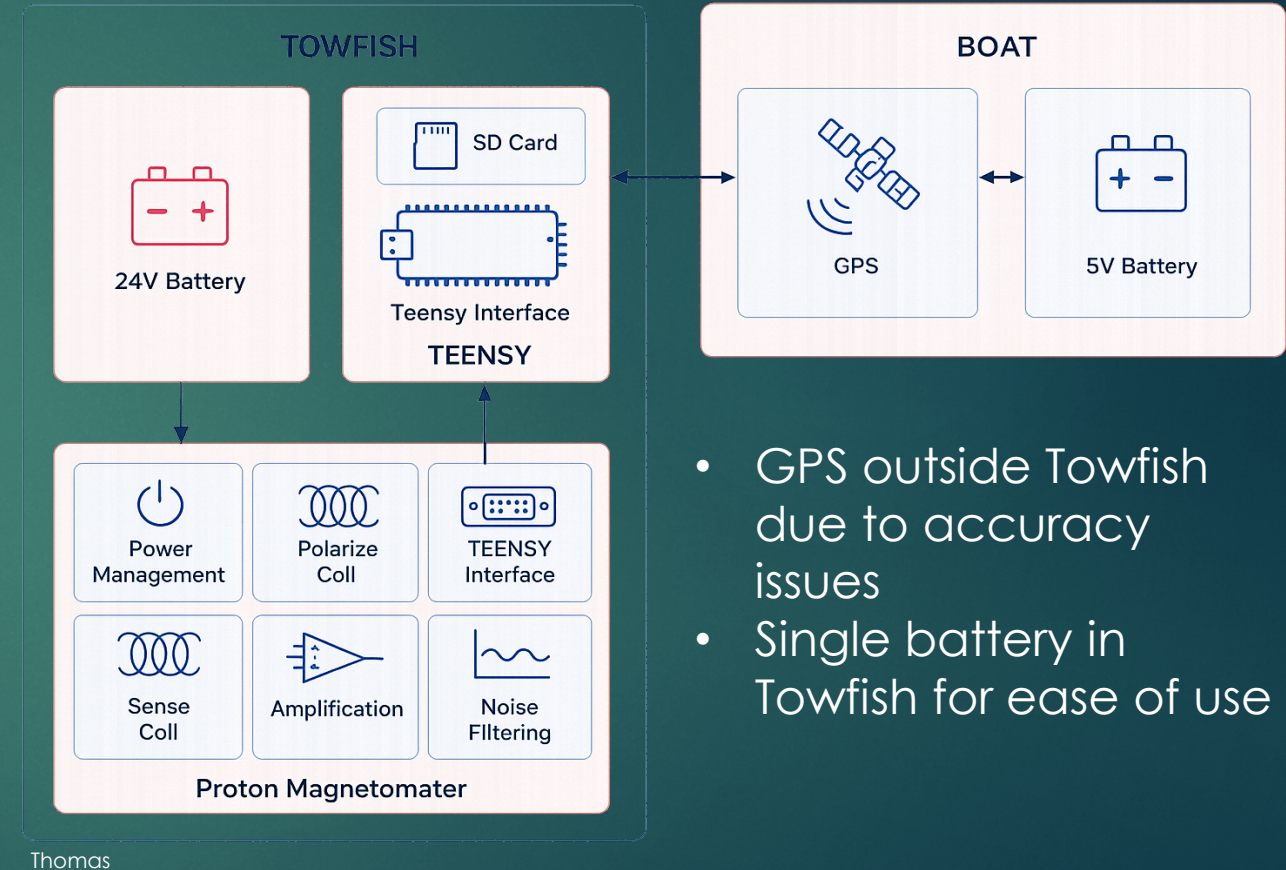


Our Design

Operating Instructions

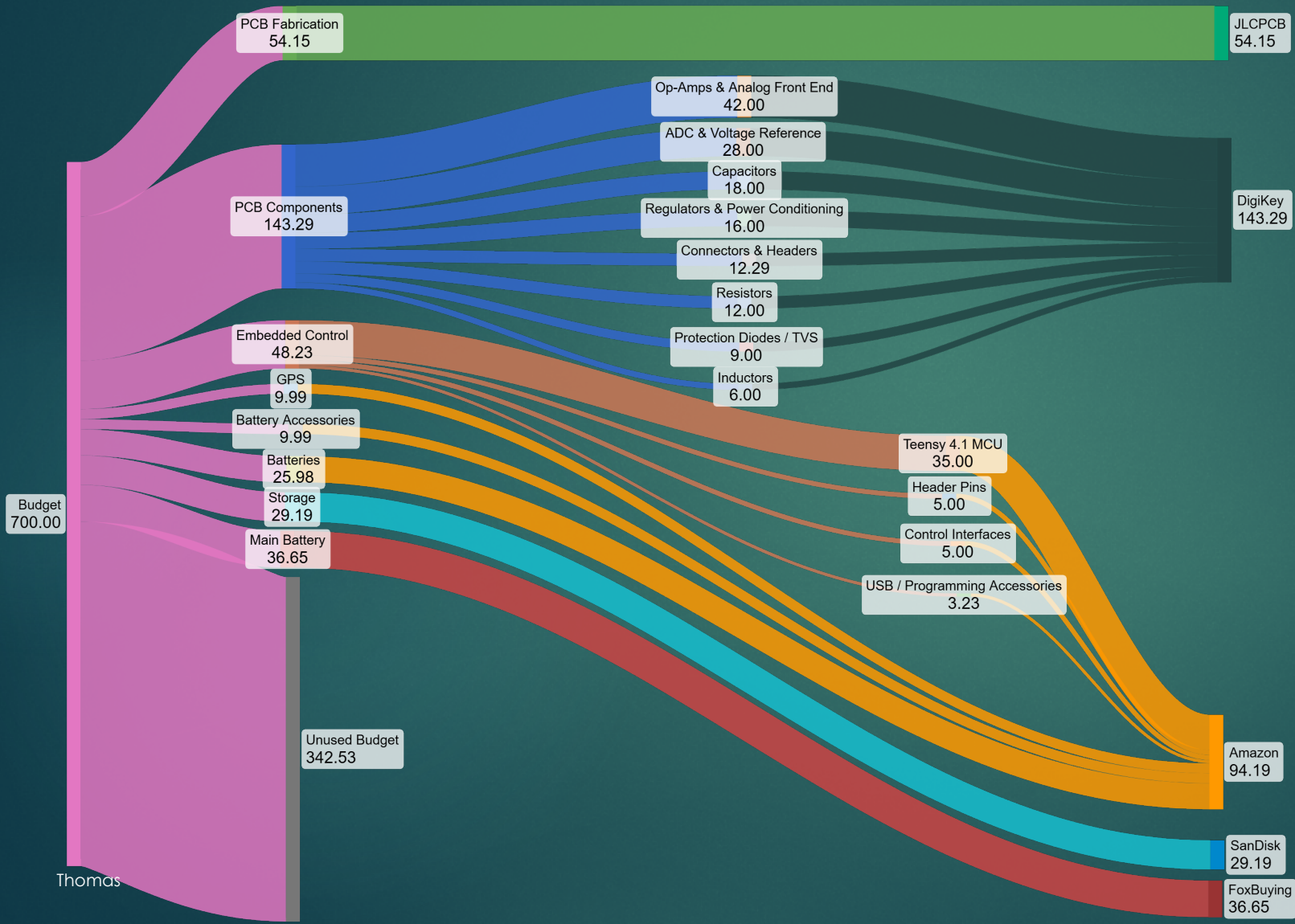
1. Ensure Teensy has SD card inserted
2. Ensure coils are securely screwed into PCB
3. Ensure gps unit attached to Teensy
4. Plug gps unit into 5V battery pack
5. Plug in 24V battery into jack on PCB
6. Move magnetometer around desired detection area
7. Unplug 24V battery and gps unit
8. Remove SD card from Teensy and insert into computer
9. Run software on recorded information

System Architecture



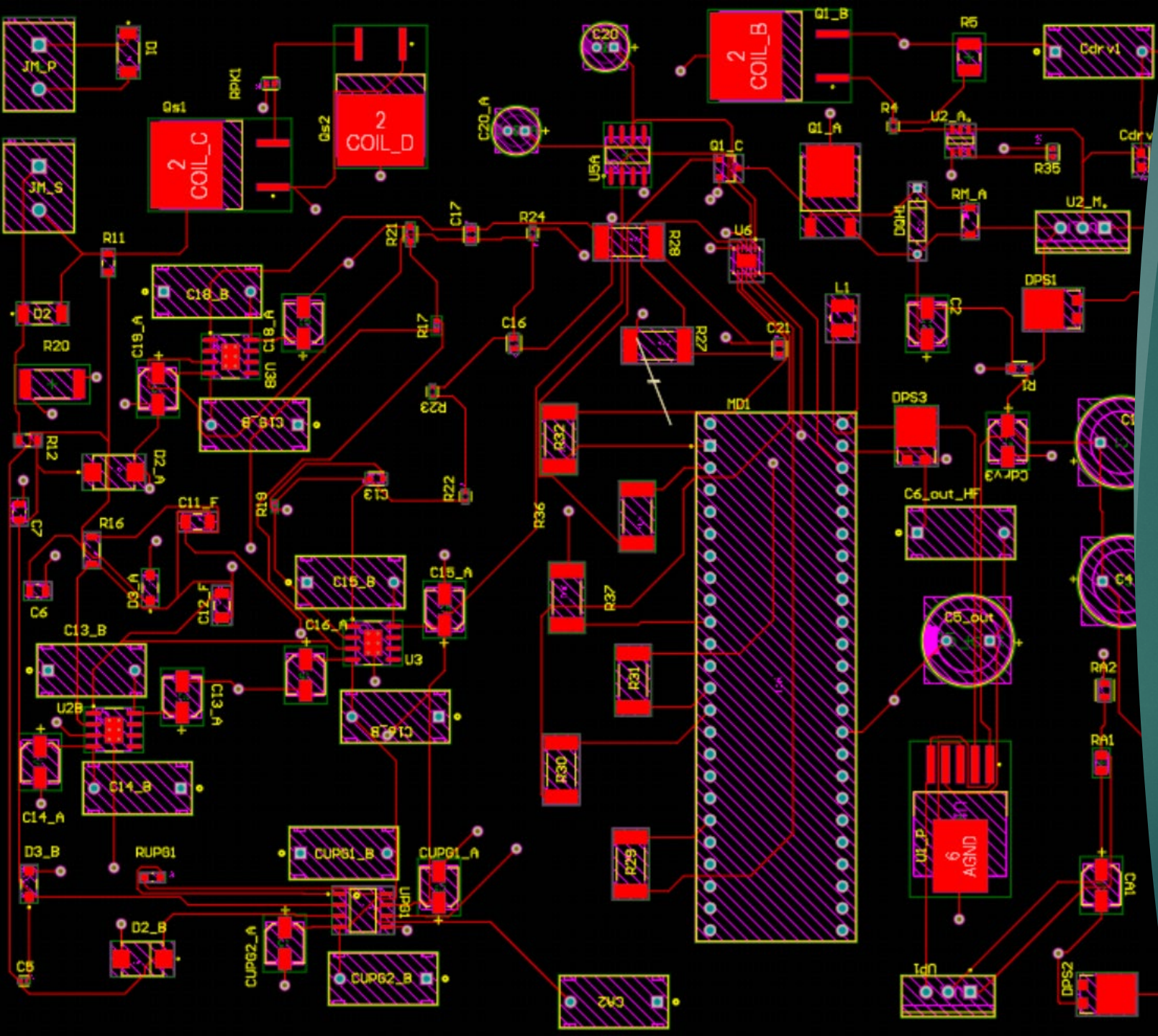
- GPS outside Towfish due to accuracy issues
- Single battery in Towfish for ease of use

Cost Breakdown



Category	Item / Part	Vendor	Qty	Unit Cost	Total Cost
PCB	DigiKey electronic	DigiKey	1 lot		\$143.29
Components	components				
PCB	Custom PCB	JLCPCB	5		\$54.15
Fabrication	manufacturing				
Embedded Control	Teensy 4.1 with Pins	Amazon	1	\$48.23	\$48.23
GPS	GT-U7 GPS Module	Amazon	1	\$9.99	\$9.99
Battery Accessories	AA battery holder bundle	Amazon	1	\$9.99	\$9.99
Batteries	Rechargeable AA batteries + charger	Amazon	1	\$25.98	\$25.98
Storage	SanDisk Ultra 32 GB microSDHC card with adapter	SanDisk	1	\$20.99	\$29.19
Main Battery	22.2/24/25.2 V 6S1P 18650 lithium battery pack	FoxBuyi ng	1	\$27.96	\$36.65
Total					\$357.47

Junwen



PCB DESIGN

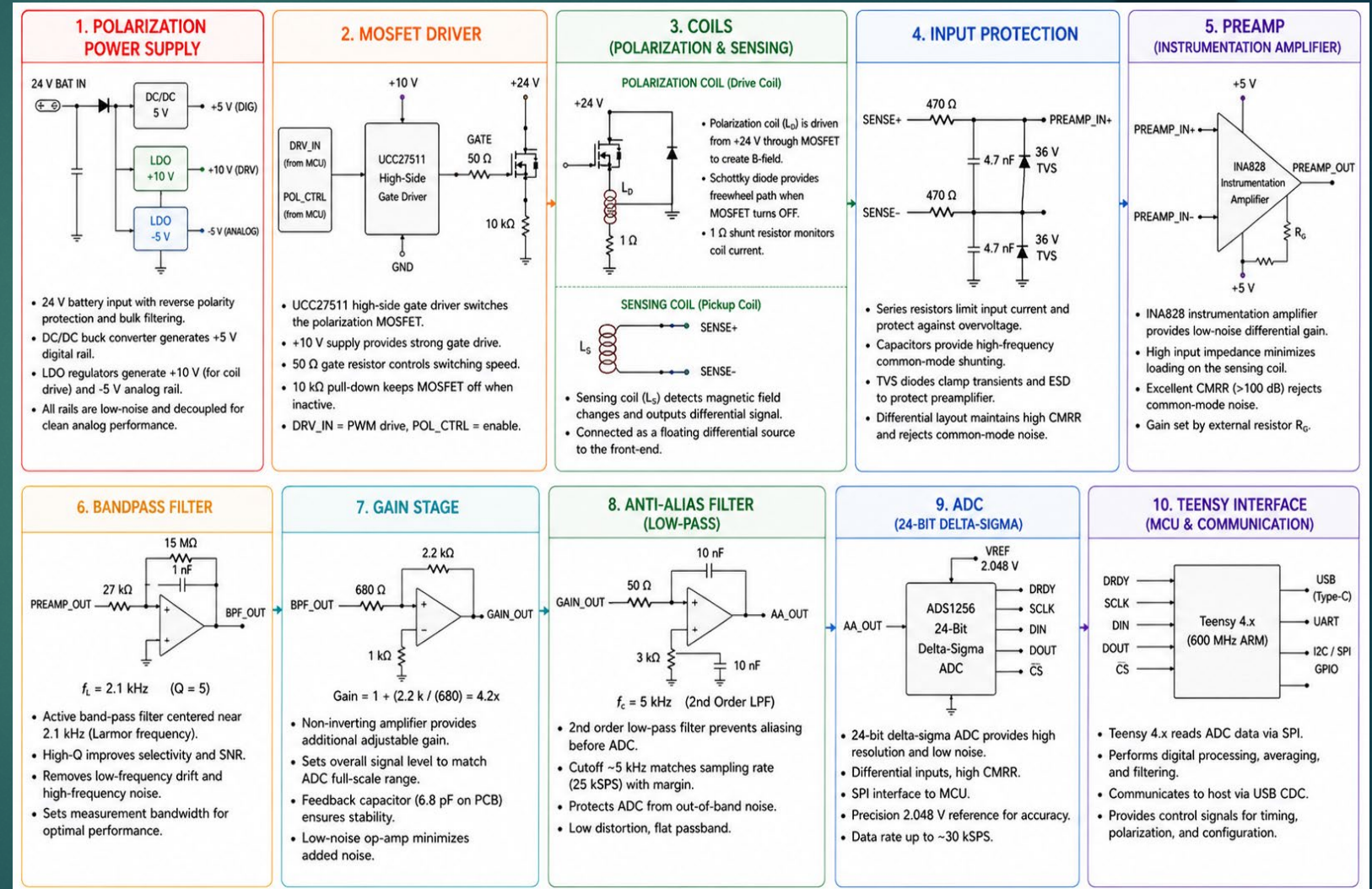
Calculations Overview

- The calculations for values needed on the PCB were done using an excel spreadsheet
- Since this was done in the beginning of the project and we were unsure about initial conditions, all calculated values(shown on future slides) utilize the parameters shown on the right; everything is dependent on them
- An example of how this would work: if the battery value was reduced, the polarization coil would have had to be smaller, the sensing coil more sensitive, the gain stage more potent, and the filtering more accurate to account for the smaller input.

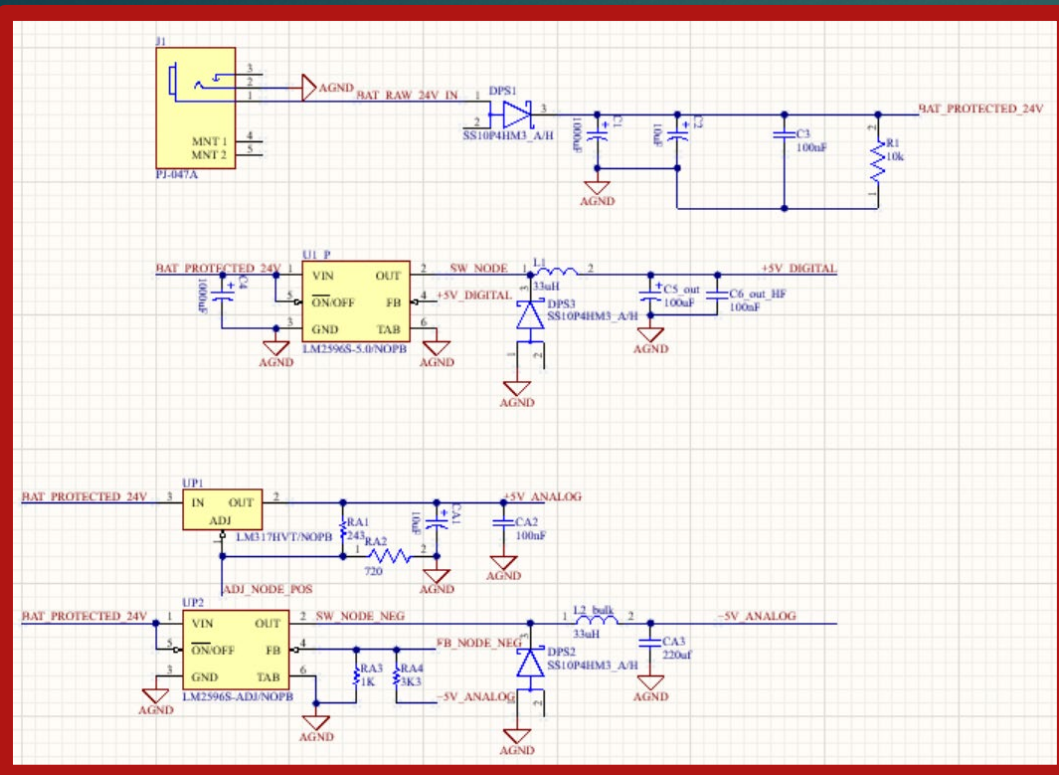
Name	Value	Units	Category
mu0	0.00000126 H/m		Global constraint
gamma_Hz_per_uT	42.576 Hz/uT		Global constraint
B_earth_uT	50 uT		Global constraint
DeltaB_uT	0.00025 uT		Global constraint
k_pol	75 × Earth field		Global constraint
T_pol_s	0.8 s		Global constraint
T_cycle_s	2.8 s		Global constraint
Temp_K	293 K		Global constraint
BW_Hz	0.5 Hz		Global constraint
T2_s	0.5 s		Global constraint
SNR_required	7.476275042 ratio		Global constraint
V_signal_min_V	0.01 V		Global constraint
V_supply_V	24 V		Power constraint
V_gate_drive_V	10 V		MOSFET-driver constraint
Qg_total_C	0.00000004 C		MOSFET-driver constraint
t_switch_target_s	0.0000002 s		MOSFET-driver constraint
V_diode_forward_V	0.7 V		Module constraint
Tj_max_C	125 °C		Thermal constraint
Tambient_C	40 °C		Thermal constraint
R_mos_ohm	0.01 ohm		MOSFET constraint
R_trace_ohm	0.02 ohm		Module constraint
V_sense_max_V	0.1 V		Module constraint
I_inrush_max_A	10 A		Module constraint
T_discharge_s	5 s		Module constraint
pol_coil_gauge	21 AWG		Coil design
pol_turns	700 turns		Coil design
pol_diameter_m	0.0762 m		Coil design
pol_radius_m	0.0381 m		Coil design
pol_length_m	0.0762 m		Coil design
pol_core	Air		Coil design
pol_inductance_H	0.002 H		Coil design
pol_resistance_ohm	20 ohm		Coil design
sensing_coil_gauge	30 AWG		Coil design
sensing_turns	2000 turns		Coil design
sensing_diameter_m	0.0762 m		Coil design
sensing_radius_m	0.0381 m		Coil design
sensing_core	Water		Coil design
sensing_inductance_H	0.06 H		Coil design
sensing_resistance_ohm	100 ohm		Coil design
ADC_fullscale_V	2 V		ADC constraint
ADC_bits_selected	18 bits		ADC constraint
ADC_sample_rate_Hz	25000 samples/s		ADC constraint
MCU_clock_Hz	600000000 Hz		MCU constraint
GPS_module	GT-U7		GPS/module constraint
GPS_update_rate_Hz	5 Hz		GPS/module constraint
GPS_power_input_V	5 V		GPS/module constraint
GPS_positional_accuracy_m	2.5 m		GPS/module constraint
GPS_velocity_accuracy_m_s	0.1 m/s		GPS/module constraint
SPI_min_clock_Hz	250000 Hz		MCU/SPI constraint
Preamp_target_out_V	0.5 V		Preamp constraint
Gain_stage_output_V	1.6 V		Gain stage constraint

Modules

- We divided our PCB into 10 distinct 'module' for ease of design and calculation
- Each of these modules represents a function of the proton magnetometer that the PCB needs to contain
- The calculations are also split between these modules, making it easier to change one thing without cascading effects
- The next 10 slides will go through each section and their accompanying calculations



Module 1 - Power



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- Contains the main system electrical constants and operating assumptions
- Uses a 24V input supply for the full hardware system
- Generates regulated voltage rails for analog and digital circuitry
- Provides stable power for the Teensy, ADC, and analog amplifiers

PCB FILE | Spreadsheet

GLOBAL CONSTANTS / POWER SUPPLY		vals1	pinouts1
mu0		0.00000126	
gamma_Hz_per_uT		42.576	
B_earth_uT		50	
Deltaf_target_Hz(change based on sensitivity)		0.010644	
k_pol		75	
T_pol_s		0.8	
T_cycle_s		2.8	
r_coil_m		0.0381	
L_coil_length_m		0.0762	
N_turns		700	
A_wire_m2	5.03E-07		
rho_cu		1.68E-08	
R_mos_ohm(.01-.03)		0.01	
R_trace_ohm		0.02	
V_sense_max_V		0.1	
I_inrush_max_A		10	
T_discharge_s		5	
DERIVED COIL + POLARIZATION VALUES			
B_pol_T		0.00375	
I_pol_A		1.2	
I_wire_m		167.5725521	
R_coil_ohm		20	
R_total_ohm		20.03	
V_pol_V		24.036	
V_supply_V		24	
MODULE 1 COMPONENTS			
C1_bulk_F		0.001	Pin: +POLARIZE_SUPPLY to PGND
R1_bleed_ohm		10000	Pin: +POLARIZE_SUPPLY to PGND
P_R1_W		0.0576	
R2_sense_ohm(moved to next module)		1	Pin: MOSFET_SOURCE to PGND
P_R2_W		1.44	
f_rep_Hz		0.357142857	Pin: +POLARIZE_SUPPLY to PGND
DeltaV_ripple_V		0.48	
C2_decouple_F		0.00001	
R3_limit_ohm(too small to use - disregard)			Pin: SUPPLY_IN to +POLARIZE_SUPPLY
P_R3_W			
ENERGY + TIME CONSTANT CHECK			
L_coil_H		0.002	
tau_s		0.0001	
T_pol_requirement		0.0005	
E_coil_J		0.00144	
P_avg_W		0.000514286	
SENSITIVITY LINK			
DeltaB_uT		0.00025	

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Datasheets:

J1 (PJ-047A): CUI Devices DC Power Jack

U1_P (LM2596S-5.0/NOPB): TL 5.0V Buck Regulator

UP1 (LM317HVT/NOPB): TI High Voltage Positive Linear Regulator

UP2 (LM2596S-ADJ/NOPB): TI Adjustable Buck Regulator



- Thomas

MOSFET DRIVER	vals2	pinouts2
V_gate_drive_V	10	
Qg_total_C	0.00000004	
t_switch_target_s	0.0000002	
V_diode_forward_V	0.7	
Tj_max_C	125	
Tambient_C	40	
MOSFET ELECTRICAL REQUIREMENTS		
MOSFET_Vds_required_V	36	
MOSFET_Id_required_A	4.5	
MOSFET_P_conduction_W	0.0144	
MODULE 2 COMPONENTS		
GATE DRIVE RESISTOR		
I_gate_required_A	0.2	Pin: GATE
R4_gate_ohm	50	
P_R4_W	1.42857E-07	
GATE PULL-DOWN RESISTOR		
R5_pulldown_ohm	10000	Pin: GATE to SOURCE
P_R5_W	0.01	
D1_flyback		
D1>If_peak_A	3	Across: COIL_A to COIL_B
D1_Vr_required_V	36	
D1_energy_J	0.00144	
RC SNUBBER		
C3_snub_F(can be removed) or 1-10nF	0.0000001	Pins: DRAIN–SOURCE
R6_snub_ohm		
P_R6_W		
SWITCHING LOSS ESTIMATION		
P_switching_W	1.02857E-06	
TOTAL MOSFET POWER		
MOSFET_P_total_W	0.014401029	
THERMAL CONSTRAINT		
MOSFET_Rth_required_C_per_W	5902.356181	

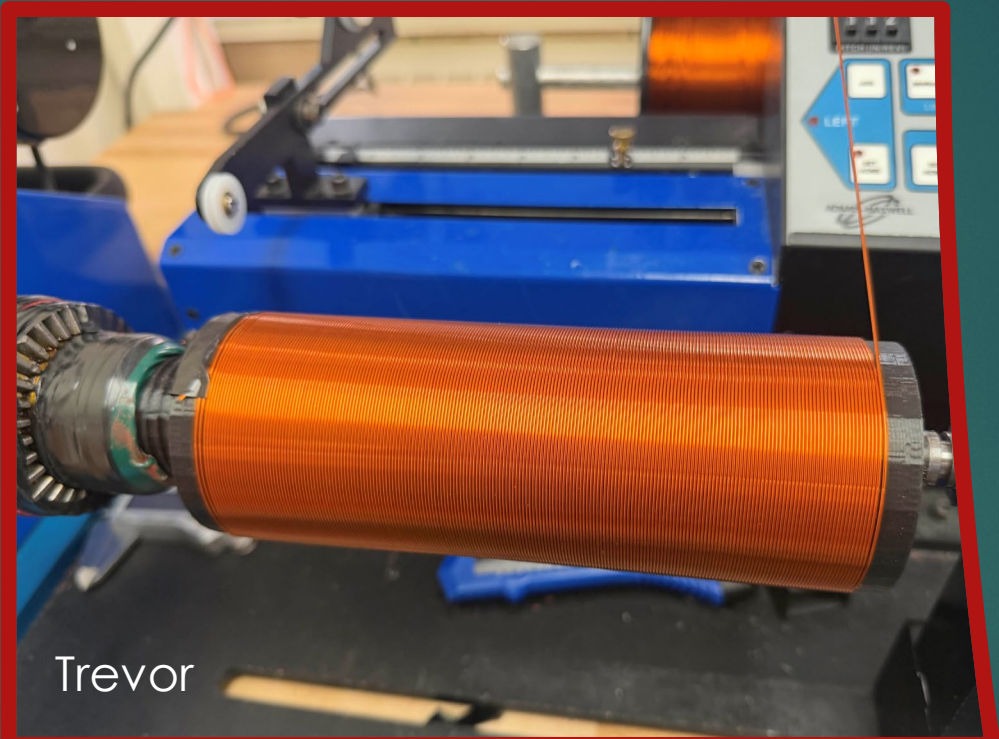
U2_M (LM317HVT/NOPB): [TI High Voltage Positive Linear Regulator](#)
U2_A (UCC27511DBVR): [TI Single-Channel High-Speed Gate Driver](#)
Q1_B (IPB017N08N5): [Infineon OptiMOS Power MOSFET](#)



Module 3 – Coils



Trevor

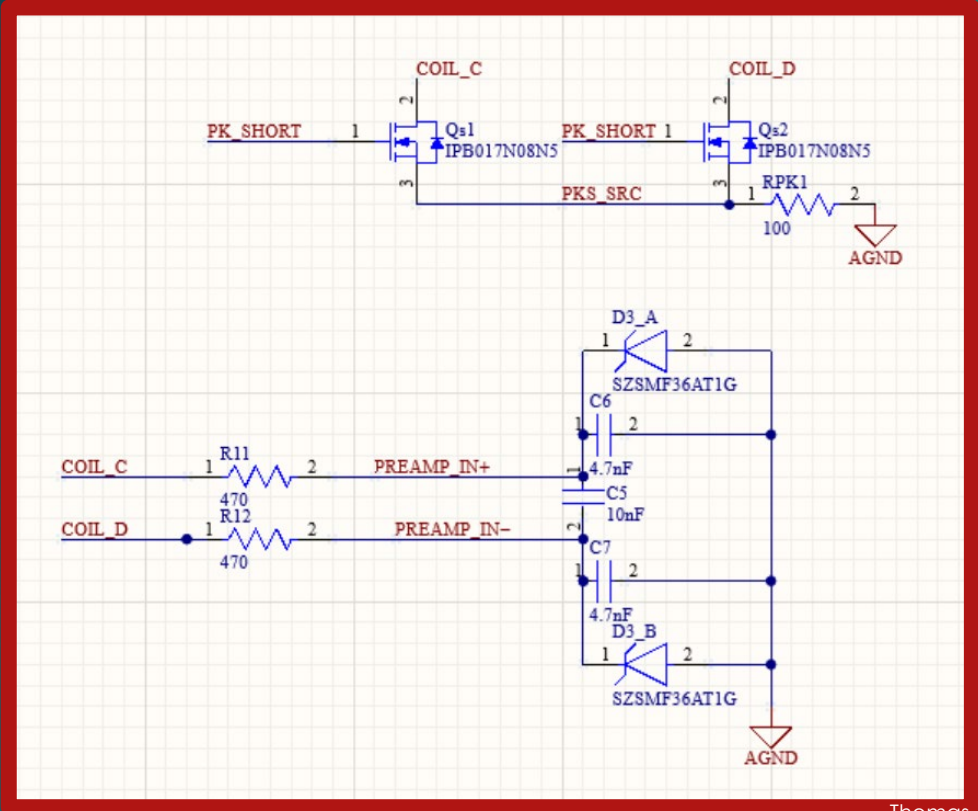


Trevor

- Coils discussed in much greater detail in [Coil Design Section](#)

POLARIZATION COIL		vals3
r_pol_coil_m		0.0381
diameter_pol_coil_m		0.0762
B_pol_T		0.00375
Deltaf_target_Hz		0.010644
T_meas_s		2
Temp_K		293
BW_Hz		0.5
T2(sig_decay)		0.5
FID AMPLITUDE REQUIREMENT		
SNR_required		7.476275042
V_signal_min_V		0.01
V_coil_required_V		3.04989E-05
COIL SIGNAL MODEL		
omega0_rad		13375.64488
V_coil_estimate_V		3.04989E-05
N_turns_required		700
WIRE LENGTH + RESISTANCE		
l_wire_m		167.5725521
wire_gauge		21
coil_core	Air	
R_pol_coil_ohm		20
INDUCTANCE		
L_pol_coil_H		0.002
THERMAL NOISE		
Vn_rms_V		8.99478E-10
SNR_actual		33907.37028
POLARIZATION CURRENT UPDATE		
L_pol_A		1.2
TIME CONSTANT CHECK		
tau_s		0.0001
T_pol_requirement		0.0005
Q FACTOR		
f0_Hz		2128.8
Q_factor		8.025386929
SENSING COIL		vals4
sensing_coil_gauge		30
sensing_turns		2000
sensing_diameter_m		0.0762
sensing_core	Water	
sensing_inductance_H		0.06
sensing_resistance_ohm		100

Module 4 – Input Protection

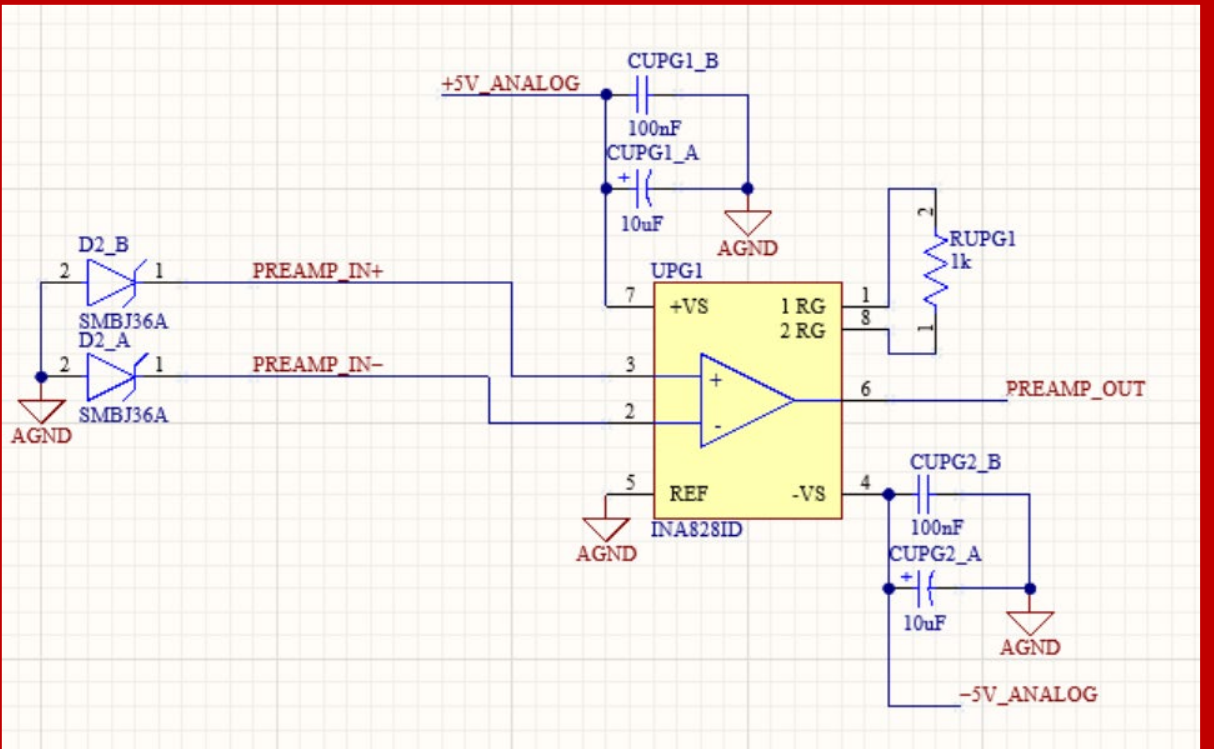


- Protects sensitive components from voltage transients
- Prevents high-current spikes from damaging the preamp stage
- Uses clamping and limiting components for signal safety
- Reduces the effect of switching noise from the polarization circuit
- Helps maintain stable amplifier input conditions
- Prevents accidental overvoltage conditions at the signal input

[PCB FILE](#) | [Spreadsheet](#)

INPUT PROTECTION	vals5	pinouts5
R_coil_ohm	100	
L_coil_H	0.06	
f0_Hz	2128.8	
BW_Hz	0.5	
V_preamp_max_V		
		DIFFERENTIAL DAMPING RESISTOR
R10_diff_ohm (remove unless need specific damping)	31830988.62	RX+ to RX- (across inputs)
		R11_in_plus
I_preamp_limit_A	0.005	Pin: RX+ → PREAMP_IN+
R11_in_plus_ohm	470	
P_R11_W	2.12766E-07	
R12_in_minus_ohm	470	
		C5_diff
f_diff_cutoff_Hz	31830988.62	Pin: PREAMP_IN+ to PREAMP_IN-
C5_diff_F	0.00000001	
		C6_cm_plus
Xc_target_ohm	1.063829787	Pin: PREAMP_IN+ → AGND
C6_cm_plus_F	4.7E-09	
C7_cm_minus_F	4.7E-09	
D3_A_ESD_plus	SZSMF36AT1G	D3_ESD_plus
D3_B_ESD_minus	SZSMF36AT1G	Pin: PREAMP_IN+ → AGND
D2_B_TVS_plus	SMBJ36A	
D2_A_TVS_minus	SMBJ36A	
INPUT NOISE CONTRIBUTION		
Vn_R11_V	1.95002E-09	
Vn_total_input_V	2.75775E-09	
SNR_post_protection	3626148.819	

Module 5 – Preamp



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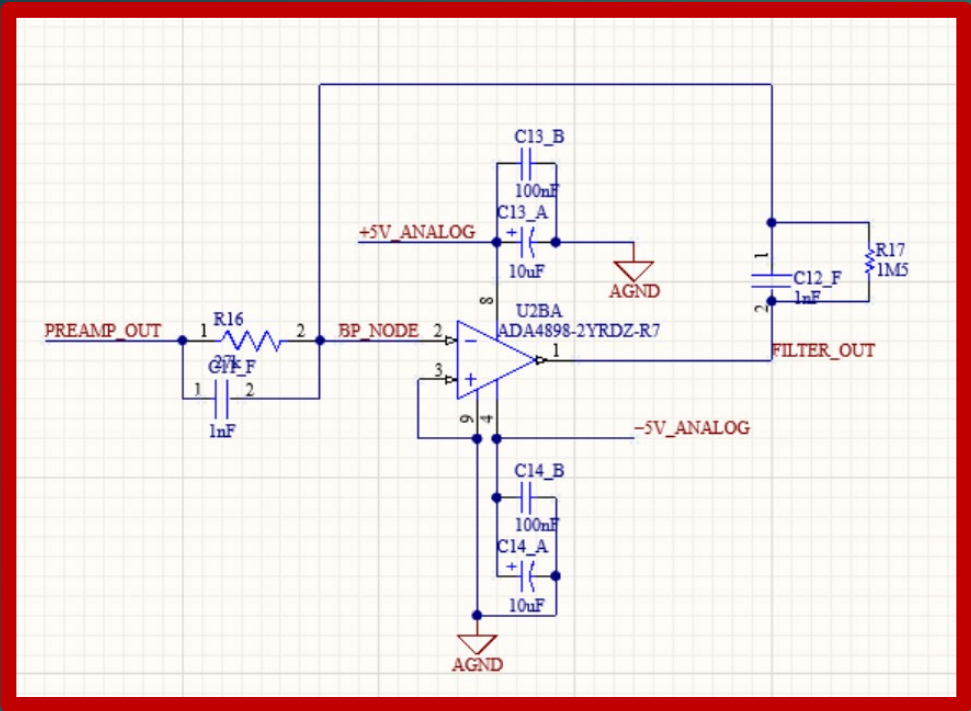
- Amplifies the weak sensing coil output signal
- Provides the first stage of low-noise analog amplification
- Increases signal amplitude before filtering stages
- Improves overall signal-to-noise ratio of the system
- Uses analog op-amp circuitry powered from regulated rails
- Interfaces directly with the sensing coil output
- Conditions the signal for the bandpass filter stage

PREAMP	vals6	pinouts6
V_coil_required_V	3.04989E-05	
V_target_out_V	0.5	
f0_Hz	2128.8	
BW_Hz	0.5	
Vn_input_total_V	2.75775E-09	
REQUIRED GAIN		
Gain_required	16394.01945	
UPG1_INA828ID	INA828ID	U1 Preamp
RUPG1_gain_ohm		1000Pin: IN- to AGND → R14_gain
Gain_INA828_nominal		50.4Pin: OUT to IN- → R13_feedback
		R15_bias
R15_bias_ohm		1000Pin: IN+ → AGND
CUPG1_A_plus_decouple_F	0.00001	
CUPG1_B_plus_decouple_F	0.0000001	C8_feedback
CUPG2_A_minus_decouple_F	0.00001	Pin: OUT to IN-
CUPG2_B_minus_decouple_F	0.0000001	
PREAMP OUTPUT NOISE		
Vn_preamp_out_V	1.3899E-07	
OP-AMP INPUT NOISE REQUIREMENT		
Vn_opamp_max_V	2.37894E-13	
en_opamp_max_V_per_rHz	1.36E-09	
GAIN BANDWIDTH REQUIREMENT		
GBW_required_Hz	697991772.2	
OUTPUT SWING REQUIREMENT		
Vout_peak_required_V	0.5	
	C9_decouple_plus + C10_decouple_minus	
C13_A_decouple_plus_F	0.00001	Pin: +ANALOG_SUPPLY → AGND
C13_B_decouple_plus_F	0.0000001	Pin: -ANALOG_SUPPLY → AGND
C14_A_decouple_minus_F	0.00001	
C14_B_decouple_minus_F	0.0000001	

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Datasheet:
UPG1 (INA828ID): [TI High-Precision Instrumentation Amplifier](#)

Module 6 – Bandpass Filter



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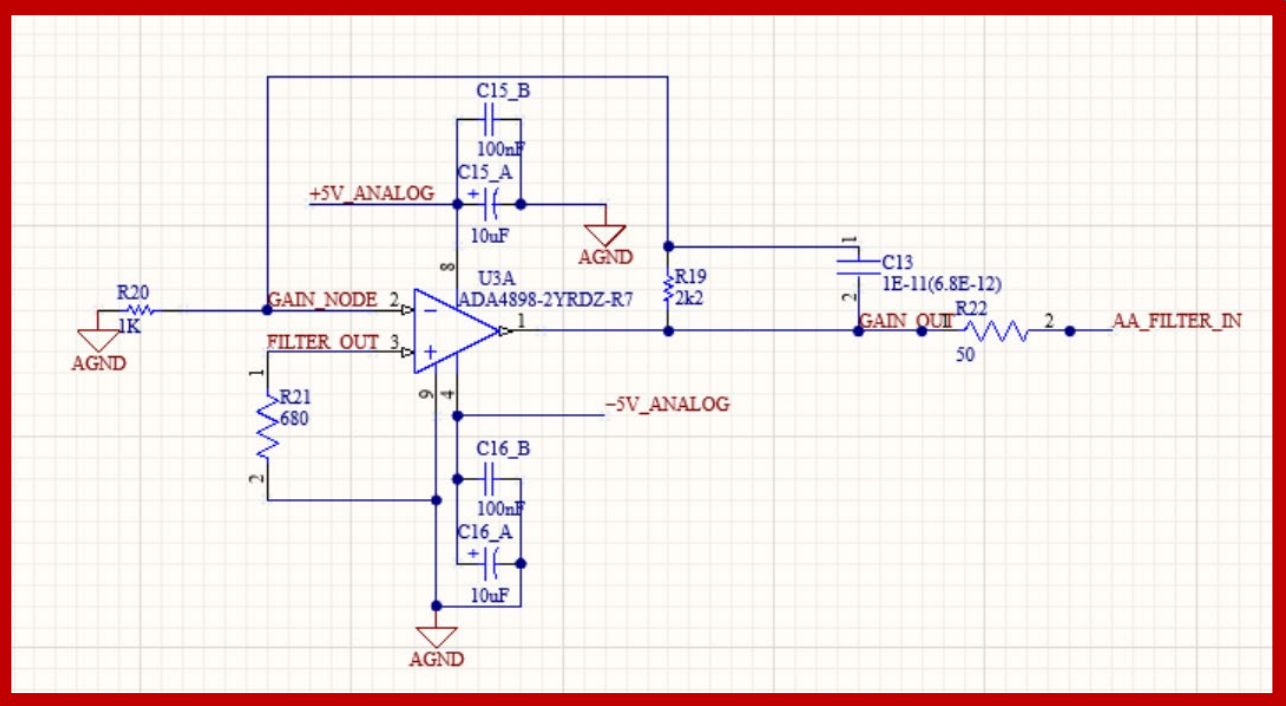
- Passes only the desired NMR frequency range
- Rejects low-frequency drift and DC offsets
- Removes unwanted high-frequency electrical noise
- Defines the usable bandwidth of the receiver system
- Improves signal clarity before additional amplification
- Helps isolate the resonant signal from interference
- Ensures cleaner input to the gain stage circuitry

BANDPASS FILTER	vals7	pinouts7
f0_Hz	2128.8	
BW_Hz	200	
Gain_preamp_out	50.4	
FILTER TARGET PARAMETERS		
Q_target	10	
Gain_center	1	
SELECT BASE CAPACITOR VALUE		
U2BA_ADA4898-2	ADA4898-2YRDZ-R7	
C12_F	1.00E-09	
RESISTOR CALCULATIONS (MFB Bandpass)		
R16_ohm	27000	
R17_ohm	1500000	
R18_ohm	7476.275042	
FILTER CHECKS		
f0_check_Hz	2128.8	
BW_check_Hz	200	
THERMAL NOISE CONTRIBUTION		
Vn_R16_V	2.95599E-07	
Vn_R17_V	2.20326E-06	
Vn_filter_total_V	2.223E-06	
AFTER FILTER		
V_signal_after_filter	0.5	

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Datasheet:
U2BA (ADA4898-2YRDZ-R7): [Analog Devices High Speed, Ultra-low Noise Op-Amp](#)

Module 7 – Gain Stage



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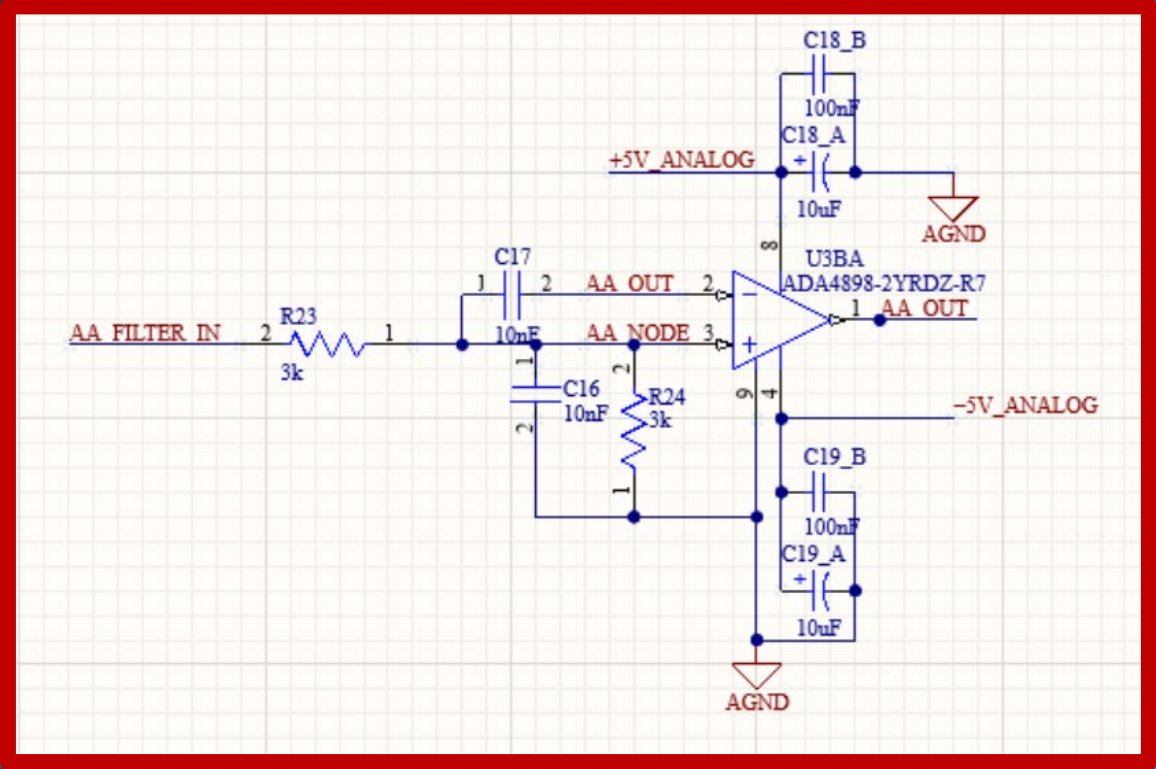
- Provides additional amplification after bandpass filtering
- Raises the signal level for accurate ADC sampling
- Sets the final analog signal amplitude before digitization
- Maintains signal integrity while increasing voltage amplitude
- Helps maximize ADC dynamic range utilization
- Interfaces directly with the anti-alias filter stage

GAIN STAGE	vals8	pinouts8
V_signal_in_V	0.5	
ADC_fullscale_V	2	
Headroom_factor	0.8	
REQUIRED OUTPUT AMPLITUDE		
V_target_ADC_V	1.6	
Gain_required_stage	3.2	
		U3 Gain Op-Amp
U3A_ADA4898-2	ADA4898-2YRDZ-R7	Pin: IN- to AGND → R20_gain
R19_feedback_ohm	2200	Pin: OUT to IN- → R19_feedback
		R21_bias
R21_bias_ohm	680	Pin: IN+ → AGND
	2200	C13_feedback
f_gain_cutoff_Hz	10638699.4	Pin: OUT → IN-
C13_feedback_F	6.8E-12	
OUTPUT NOISE		
Vn_gain_stage_out_V	7.11361E-06	
OP-AMP REQUIREMENTS		
GBW_required_Hz	136243.2	
Vout_peak_required_V	1.6	
		R22_series_out
R22_series_ohm	50	Pin: U3_OUT → AA_FILTER_IN
SUPPLY DECOUPLING		
C15_A_decouple_plus_F	0.00001	
C15_B_decouple_plus_F	0.0000001	
C16_A_decouple_minus_F	0.00001	
C16_B_decouple_minus_F	0.0000001	

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Datasheet:
U3A (ADA4898-2YRDZ-R7): [Analog Devices High Speed, Ultra-low Noise Op-Amp](#)

Module 8 – Anti Alias



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- Removes unwanted high-frequency noise before the ADC
- Prevents aliasing effects during digital sampling
- Limits signal bandwidth to match the ADC sample rate
- Uses analog filtering components for final signal cleanup
- Ensures only valid frequencies reach the ADC input
- Provides a stable conditioned signal for conversion

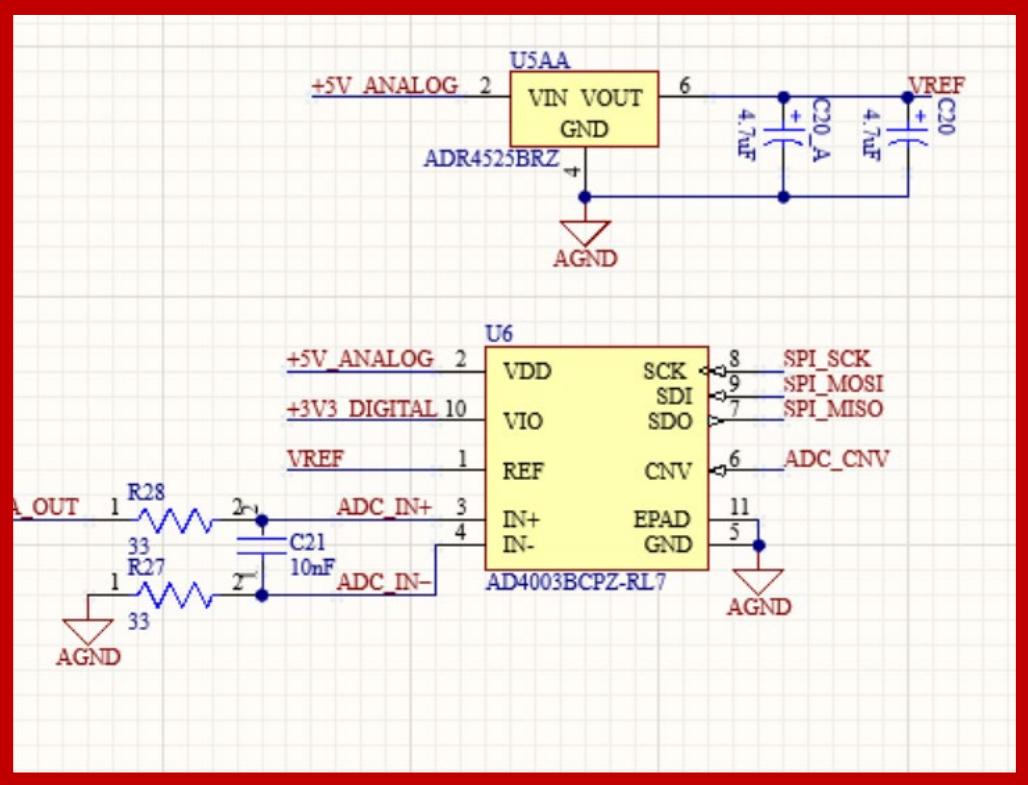
[PCB FILE](#) | [Spreadsheet](#)

ANTI-ALIAS FILTER		vals9	pinouts9
f0_signal_Hz		2128.8	
ADC_sample_rate_Hz		25000	
Gain_stage_output_V		1.6	
ANTI-ALIAS TARGET			
f_cutoff_Hz		5000	
Q_target		0.707106781	
SELECT BASE CAPACITOR			
U3BA_ADA4898-2		ADA4898-2YRDZ-R7	
C17_F		1.00E-08	
RESISTOR CALCULATION			
R23_ohm		3000	
R24_ohm		3000	
FREQUENCY CHECK			
fc_check_Hz		5305.16477	
ATTENUATION AT NYQUIST			
Nyquist_Hz		12500	
Attenuation_at_Nyquist_dB		-16.02738047	
THERMAL NOISE CONTRIBUTION			
Vn_R23_V		6.96733E-07	
Vn_AA_total_V		9.85329E-07	
TOTAL NOISE BEFORE ADC			
Vn_total_ADC_input_V		0.000001180123554	
SNR_at_ADC		222793.9209	
C18_A_decouple_plus_F		0.00001	
C18_B_decouple_plus_F		0.0000001	R25_ADC_drive
R25_ohm		50	Pin: AA_OUT → ADC_IN
C19_A_decouple_minus_F		0.00001	
C19_B_decouple_minus_F		0.0000001	
C19_decouple_minus_F		0.000001	

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Datasheet:
U3BA (ADA4898-2YRDZ-R7): [Analog Devices High Speed, Ultra-low Noise Op-Amp](#)

Module 9 – ADC



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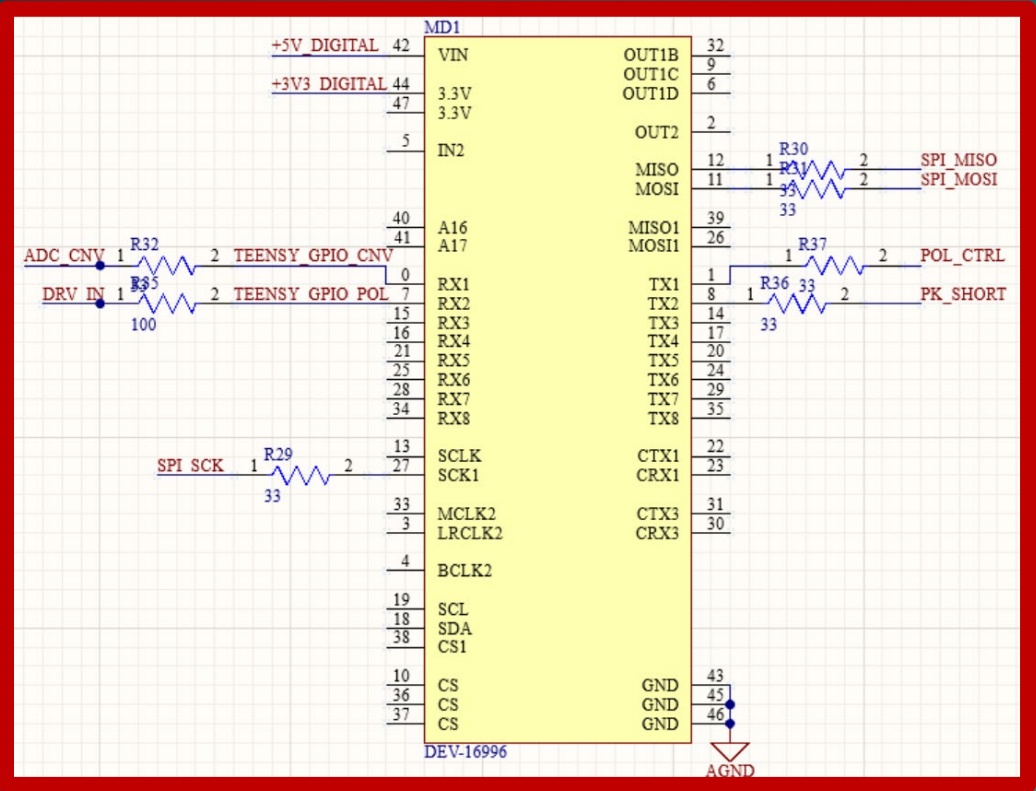
- Converts the analog receiver signal into digital samples
- Defines sampling rate and signal resolution requirements
- Produces digital output data for processing by the Teensy
- Determines measurement precision and digital accuracy

ADC	vals10	pinouts10
V_signal_peak_V	1.6	
Vn_total_ADC_input_V	8.42478E-06	
Deltaf_target_Hz	0.010644	
T_meas_s	2	
SNR_required	7.476275042	
REQUIRED ADC RESOLUTION		
V_resolution_required_V	0.000847023	
QUANTIZATION NOISE		
ADC_fullscale_V	2	
ADC_bits_selected	24	
LSB_V	1.52588E-05	
Vn_quantization_V	4.40483E-06	
TOTAL NOISE INCLUDING ADC		
Vn_total_with_ADC_V	8.42478E-06	
ADR4525BRZ_Vref_source	R26_ref_top	2.5Pin: VREF → VREF_DIV
Vref_target_V	2	
R27_ADC_IN_minus_ohm	33	
C20_A_ref_F	0.0000047 C20_ref	
C20_ref_F	0.0000047 Pin: VREF → AGND	
R28_ADC_series + C21_ADC_hold	AD4003 input RC	
R28_ADC_series_ohm	33Pin: ADC_IN → ADC_PIN	
C21_ADC_hold_F	0.00000001 Pin: ADC_PIN → AGND	
SAMPLING CLOCK REQUIREMENT		
Samples_per_cycle	50000	
Min_samples_per_period	11.74370537	
Samples_constraint	OK	

Thomas

Datasheets:
U5A (ADR4525BRZ): [Analog Devices Ultra-low Noise Voltage Reference](#)
U6 (AD4003BCPZ-RL7): [Analog Devices 24-Bit, 2 MSPS Precision SAR ADC](#)

Module 10 – Interface



Thomas

- Interfaces the ADC with the Teensy microcontroller
- Transfers sampled digital data for processing and analysis
- Controls switching and measurement sequencing operations

TEENSY INTERFACE	vals11	pinouts11
Deltaf_target_Hz	0.010644	Existing sensitivity target
T_meas_s	2	
f0_Hz	2128.8	
ADC_sample_rate_Hz	25000	
TIMER RESOLUTION REQUIREMENT		
Deltaf_time_limit	0.010644	
Time_constraint	OK	
GPS_module	GT-U7	
GPS_update_rate_Hz	5	
GPS_uart_output	UART	
GPS_power_input_V	5	
GPS_positional_accuracy_m	2.5	
GPS_velocity_accuracy_m_s	0.1	
SPI_min_clock_Hz	250000	
R29-32_SPI_SCK		
R29_SPI_SCK_ohm	33Pin: TEENSY_SCK → ADC_SCK	
R30_SPI_MISO_ohm	33Pin: TEENSY_MOSI → ADC_MOSI	
R31_SPI_MOSI_ohm	33Pin: ADC_MISO → TEENSY_MISO	
R32_ADC_CNV_ohm	35Pin: ADC_CNV → TEENSY_GPIO_CNV	
PULL-UP / PULL-DOWN RESISTORS		
R33_CS_pullup		
Pin: ADC_CS → 3V3		
R35_gate_series		
R35_gate_series_ohm	100Pin: TEENSY_GPIO → MOSFET_DRIVER_IN	
CONTROL TIMING RESOLUTION		
Timer_tick_required_s	14.95255008	
MCU_clock_Hz	600000000	
Timer_tick_actual_s	1.66667E-09	
Timer_constraint	OK	
DATA THROUGHPUT		
Samples_per_measurement	50000	
Data_rate_bytes_per_s	50000	

Datasheet:
MD1 (DEV-16996): [SparkFun Teensy 4.1 Development Board](#)

Thomas

Checks

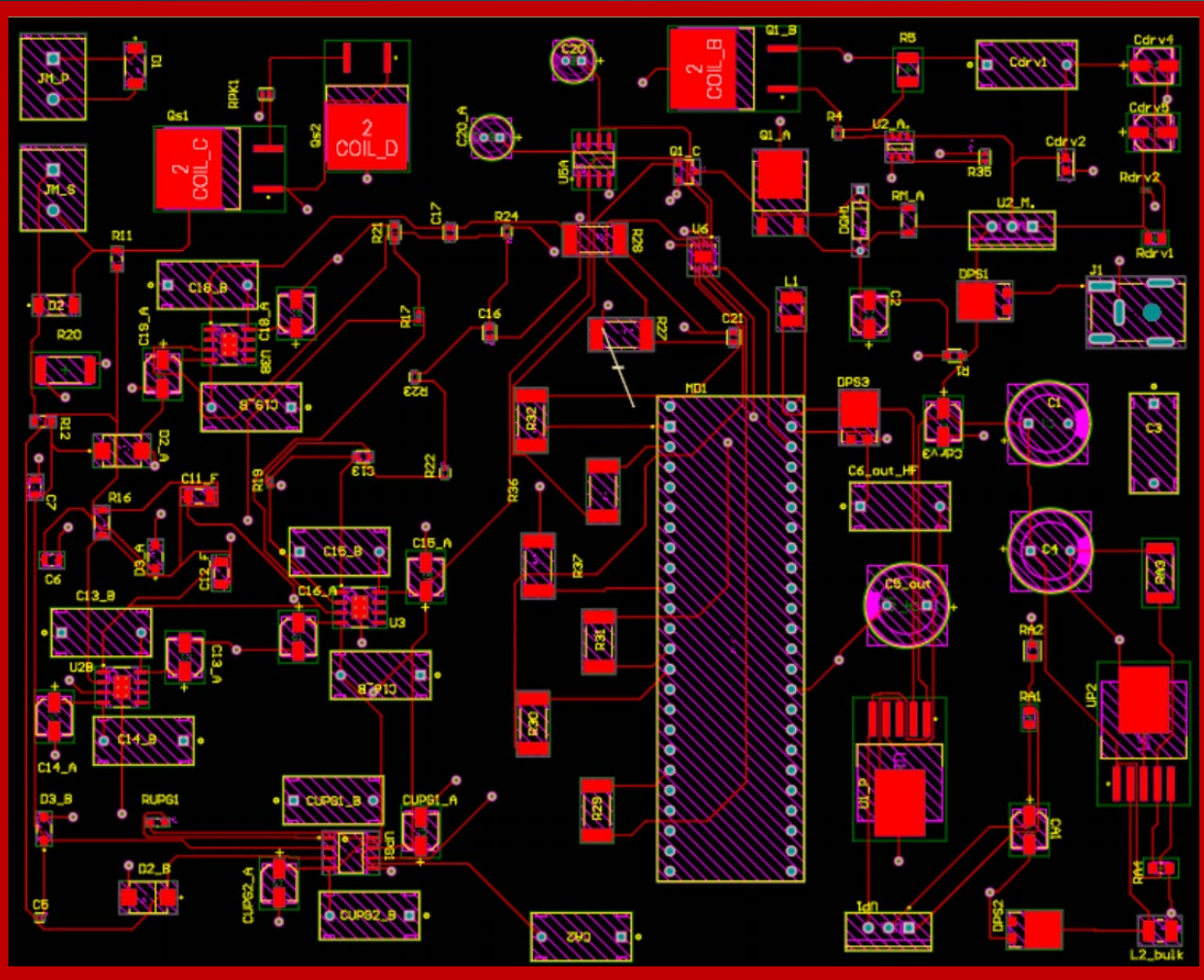
- The schematics were made while calculating all the values – hence the pinouts given in the calculation tables – these pinouts align with the actual nets in the PCK – allowing for easy reference
- A series of tests are performed on the calculated values ensuring that the design meets the desired constraints – with a focus on the required sensitivity of <1nT
- Some checks also ensure that calculated values work between modules – trying to ensure functionality
- Checks have three state – corresponding to the calculated values proximity to the desired value

Checks Summary			
Total checks	42		
PASS	42		
WARN	0		
FAIL	0		

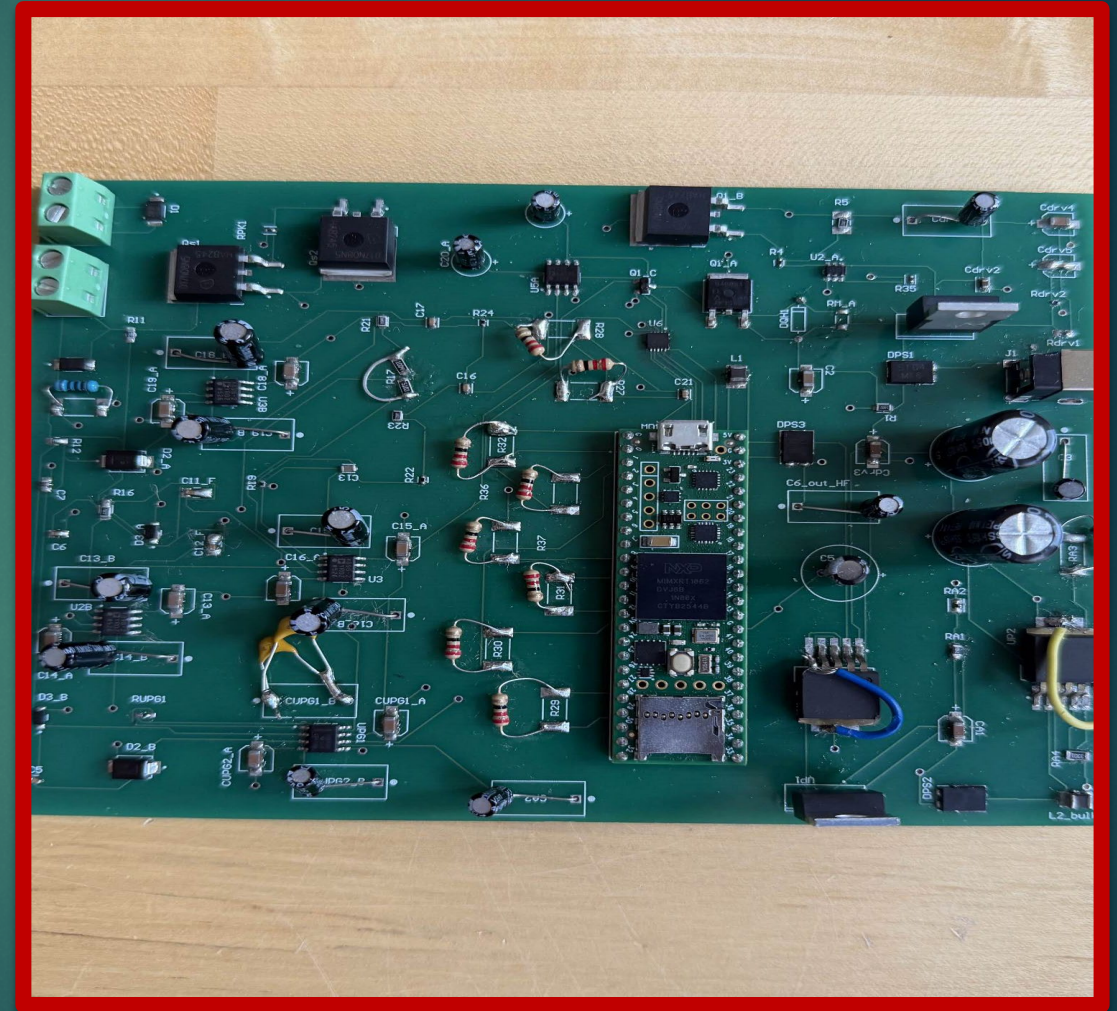
Module	Check	Calculated value	Minimum	Maximum	Unit	Status	Formula
Global	Proton Larmor frequency	2.1288E+03	1.8000E+03	2.6000E+03	Hz	PASS	$\gamma_{\text{Hz_per_uT}} \times B_{\text{earth_uT}}$
Global	Sensitivity frequency step	1.0644E-02	0.0000E+00	5.0000E-02	Hz	PASS	$\gamma_{\text{Hz_per_uT}} \times \Delta B_{\text{uT}}$
Polarization Coil	Current from netlist coil resistance	1.2000E+00	5.0000E-01	5.0000E+00	A	PASS	$V_{\text{supply}} / \rho_{\text{coil_resistance}}$
Polarization Coil	Field produced by coil	1.3890E-02	3.7500E-03		T	PASS	$\mu_0 \times N \times I / \text{length}$
Polarization Coil	Voltage required by coil	2.4036E+01	0.0000E+00	2.5200E+01	V	PASS	$I \times (R_{\text{coil}} + R_{\text{trace}} + R_{\text{mos}})$
Polarization Coil	L/R time constant	1.0000E-04	0.0000E+00	1.6000E-01	s	PASS	L/R
Polarization Coil	Polarization duty cycle	2.8571E-01	0.0000E+00	5.0000E-01	ratio	PASS	$T_{\text{pol}} / T_{\text{cycle}}$
Polarization Coil	Coil stored energy	1.4400E-03	0.0000E+00	5.0000E-01	J	PASS	$0.5 \times L \times I^2$
Sensing Coil	Turns from poster	2.0000E+03	1.9000E+03	2.1000E+03	turns	PASS	Poster coil design
Sensing Coil	Resistance from poster	1.0000E+02	8.0000E+01	1.2000E+02	ohm	PASS	Poster coil design
Sensing Coil	Inductance from poster	6.0000E-02	4.0000E-02	8.0000E-02	H	PASS	Poster coil design
Sensing Coil	Reactance at Larmor frequency	8.0254E+02	7.0000E+02	9.0000E+02	ohm	PASS	$2\pi fL$
Sensing Coil	Q at Larmor frequency	8.0254E+00	5.0000E+00	2.0000E+01	ratio	PASS	XL / R
Sensing Coil	Thermal noise over bandwidth	8.9948E-10	0.0000E+00	1.3376E-03	V rms	PASS	$\sqrt{4kTRB}$
Input Protection	R11/R12 power at min signal	2.1277E-07	0.0000E+00	1.2500E-01	W	PASS	V^2/R for 470 ohm series resistors
Input Protection	Differential RC cutoff	1.6931E+04	4.0000E+03	4.0000E+04	Hz	PASS	$1/(2\pi(R11+R12)C5)$
Input Protection	Common-mode RC cutoff	7.2048E+04	1.0000E+04	2.0000E+05	Hz	PASS	$1/(2\pi R11C6)$
Preamp	INA828 nominal gain	5.0400E+01	4.0000E+01	6.0000E+01	V/V	PASS	$1 + 49.4k/RG$
Preamp	Preamp output at min signal	5.0400E-01	0.0000E+00	2.5000E+05	V peak	PASS	$V_{\text{signal_min}} \times \text{INA828 gain}$
Bandpass	Center frequency check	2.1288E+03	1.8000E+03	2.6000E+03	Hz	PASS	Nominal Larmor frequency
Gain Stage	Non-inverting gain	3.2000E+00	3.0000E+00	3.4000E+00	V/V	PASS	$1 + R19/R20$
Anti-alias	Unity Sallen-Key cutoff estimate	5.3052E+03	4.2576E+03	5.6250E+03	Hz	PASS	$1/(2\pi\sqrt{(R23R24C16C17)})$
ADC	ADC LSB size	7.6294E-06	0.0000E+00	1.0000E-04	V/count	PASS	$\text{Fullscale} / 2^{\text{bits}}$
ADC	Sampling Nyquist margin	5.8719E+00	5.0000E+00	9.9900E+02		PASS	$fs/(2f_0)$
SPI/MCU	Minimum SPI clock for ADC readout	4.5000E+05	2.5000E+05		Hz	PASS	$\text{sample rate} \times \text{ADC bits}$
GPS	GPS update rate	5.0000E+00	1.0000E+00	1.0000E+01	Hz	PASS	Poster GPS info
MOSFET Driver	Gate drive current	2.0000E-01	2.0000E-01		A	PASS	$V_{\text{gate}} / R4$ vs Q_g/t
MOSFET Driver	MOSFET Vds recommended rating	6.0000E+01	6.0000E+01		V	PASS	$\text{supply} \times \text{margin}$
Global	Earth-field polarization multiplier	75	70	80		PASS	k_{pol} should remain near 75x Earth field
Global	Measurement bandwidth vs T2 limit	0.318309886	0	0.5	Hz	PASS	$1/(2\pi T2)$ should fit within measurement BW
Polarization Coil	Coil current density estimate	2.3857E+06	0.0000E+00	6.0000E+06	A/m²	PASS	$L_{\text{pol}} / \text{copper cross-sectional area}$
Polarization Coil	Supply current headroom	8.3333E+00	1.5000E+00			PASS	$L_{\text{inrush_max}} / L_{\text{pol}}$
MOSFET Driver	Gate charge delivered in target time	1.0000E+00	1.0000E+00			PASS	$\text{Available drive current} \times \text{target switch time} / Q_g$
MOSFET Driver	Vds margin ratio	2.5000E+00	2.5000E+00			PASS	Selected Vds rating / supply voltage
Preamp	Preamp output vs target	1.0080E+00	8.0000E-01	1.2000E+00		PASS	Output at min signal / preamp target output
Gain Stage	ADC headroom at target output	8.0000E-01	0.0000E+00	9.0000E-01		PASS	Gain stage output / ADC full-scale
Anti-alias	Cutoff-to-signal margin	2.4921E+00	2.0000E+00			PASS	Anti-alias cutoff / f_0
Anti-alias	Cutoff-to-Nyquist fraction	4.2441E-01	0.0000E+00	4.5000E-01		PASS	Anti-alias cutoff / Nyquist
ADC	Quantization-noise SNR margin	2.2702E+03	7.4763E+00			PASS	$\text{Min signal} / \text{quantization noise}$
ADC	Total ADC-input noise SNR margin	1.1870E+03	7.4763E+00			PASS	$\text{Min signal} / \text{total noise incl. ADC}$
SPI/MCU	MCU clock to SPI throughput margin	1.3333E+03	1.0000E+01			PASS	$\text{MCU clock} / \text{minimum ADC serial throughput}$
Thermal	MOSFET thermal headroom	8.5000E+01	4.0000E+01		°C	PASS	$TJ_{\text{max}} - \text{ambient}$

Thomas

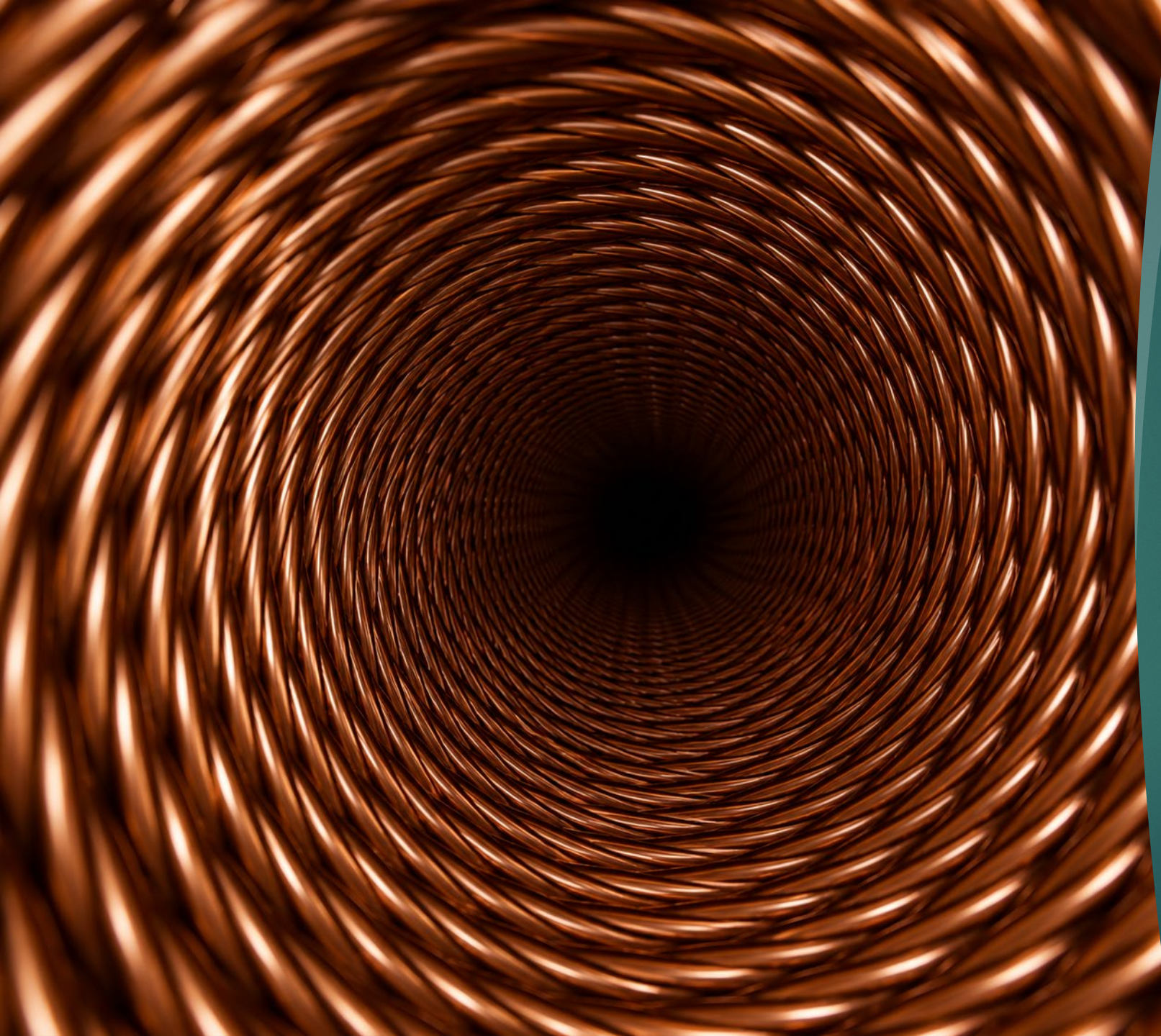
Finalized PCB Schematic and PCB based on above



Thomas



Trevor + Thomas

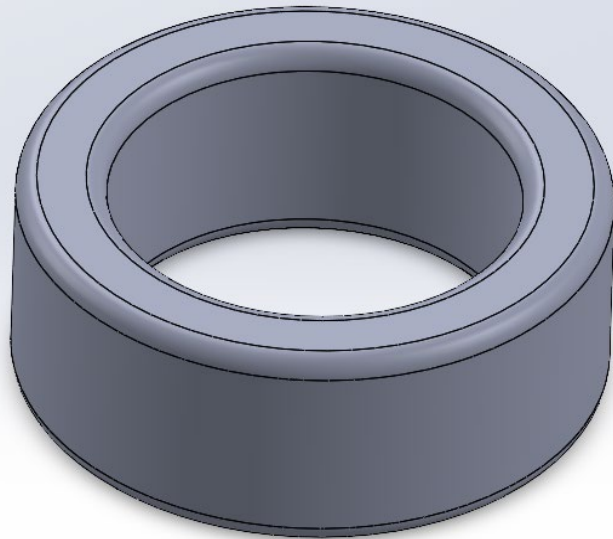


Coil Design

Original Coil Holder CAD Designs

Original Toroidal Donut Designs

- Started with a toroidal coil design after 3d printing and trying to implement ran into several issues.
- First design was not round enough so it had its own issues with winding on top of the spool issues
- To wrap the coil around it required bringing the entire spool through the hole and wasn't logical.
- There was also issues with plugging the hole after filling with proton fluid.



Trevor

[CAD FILE](#)

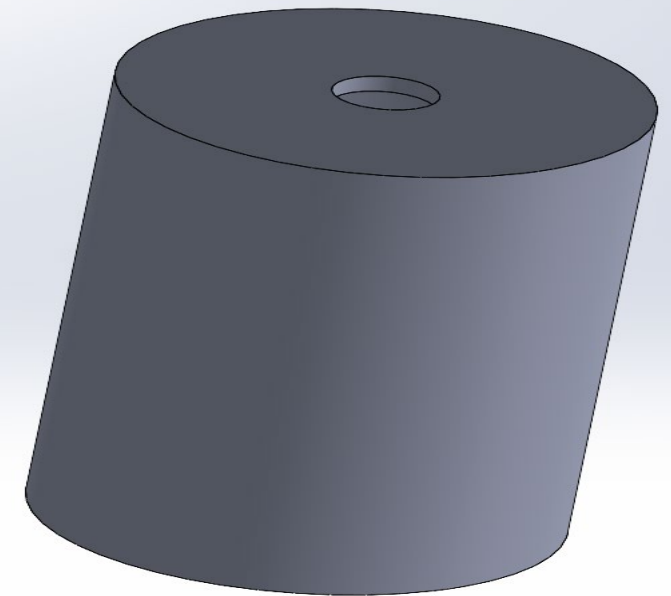


Trevor

[CAD FILE](#)

Original Solenoidal Design

- After moving on from the toroidal design the solenoidal design was introduced
- The first version of this proved to be to have issues of its own starting with the size it was created too large to fit in our tow-fish as well as made it harder to wrap in coil
- Also, it had a similar issue to the toroidal with an issue of leaking with out a better design to plug the hole after filling with proton fluid
- This design was crucial in the process of finding the final implementation



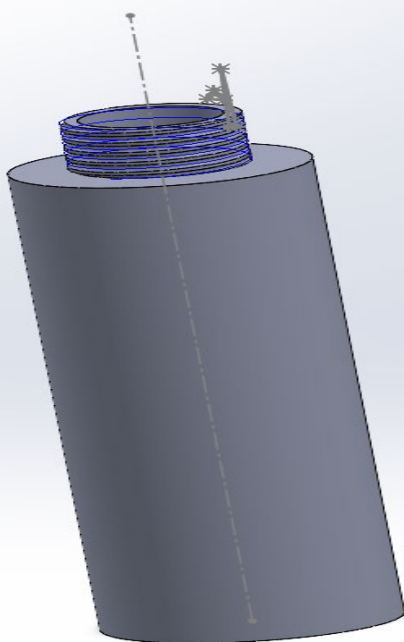
Trevor

[CAD FILE](#)

Final Coil Holder CAD Design

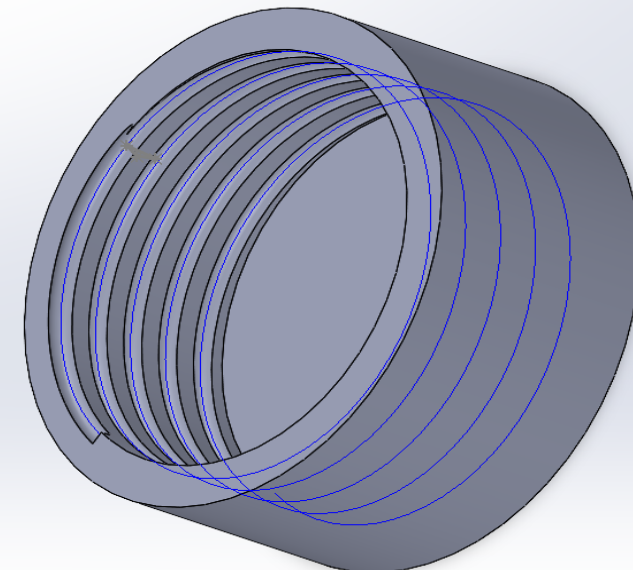
New Solenoidal Bottle Design

- This design includes new dimensions that allowed us to fit not just the 1 like previously but both of our coil bottles inside the tow-fish dimensions we were planning for.
- This new design allowed for me to create a custom thread design on both the neck of the bottle that went down to the top of the bottle for a secure design
- These same custom threads were added to the cap as well. Once the cap was tightened onto the neck of the bottle this created a pressure seal fixing the issue of leaking proton fluid.
- This design also made it way easier to wrap coil around since no longer needed to move the spool of coil and was a lot easier to fit on the coil winding machine
- The final design consisted of a 3in diameter base with a 7in length body.



Trevor

[CAD FILE](#)



Trevor

[CAD FILE](#)

Sensing Coil Winding

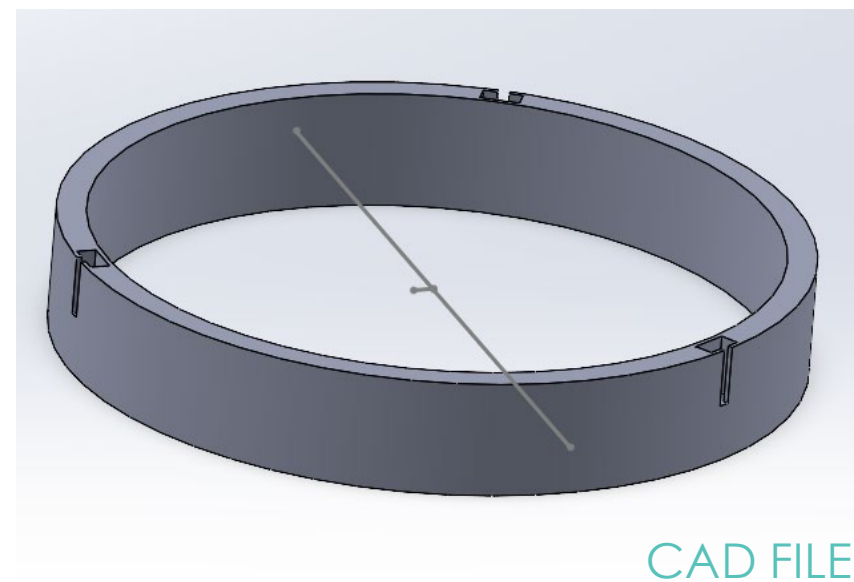
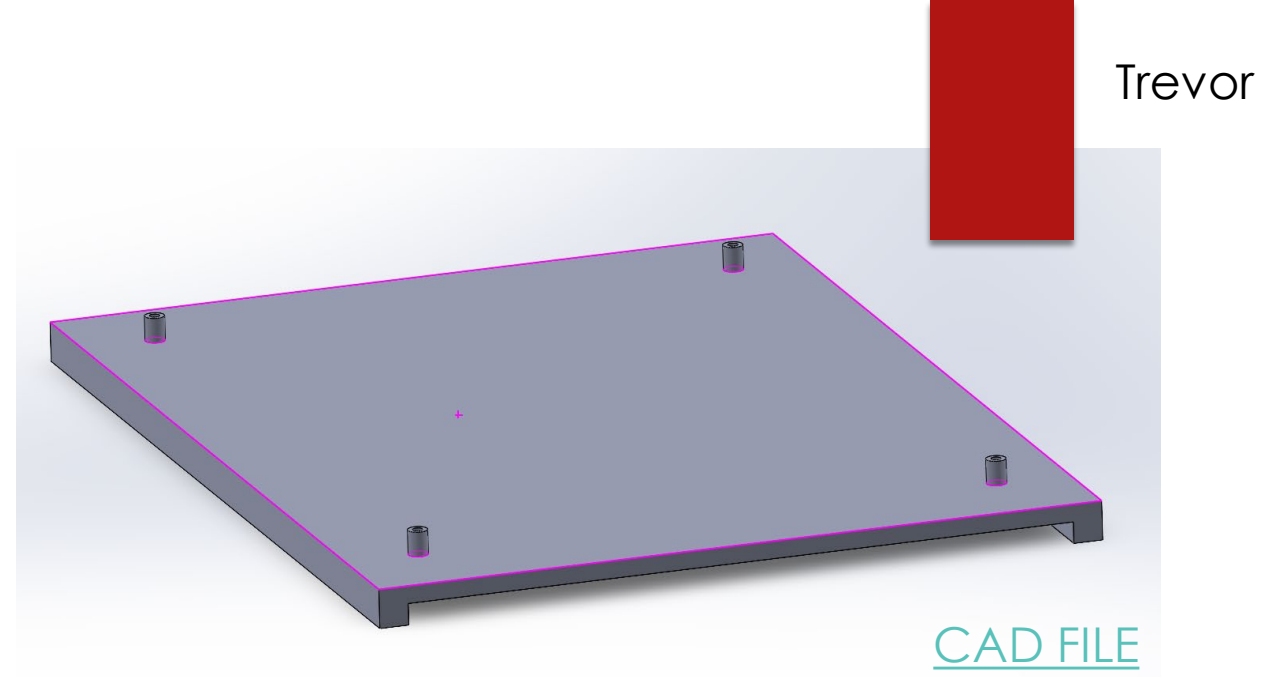
- ▶ Sensing Coil Bottle:
- ▶ This bottle holds the proton fluid which we have chosen to be distilled water.
- ▶ To hold the proton fluid efficiently without leaks the bottle is created out of a water-resistant filament PETG.
- ▶ This water is providing hydrogen protons that the magnetometer measures.
- ▶ The coil winding around the outside of the bottle acts as a detector.
- ▶ This coil is created with 30-gauge wire of about 1500-2000 turns. 30-gauge was chosen because of its thinness allows to for more turns since so many were needed
- ▶ This many turns was necessary because more turns means that more voltage is induced allowing a signal to be detected easier
- ▶ After the polarizing field switches off, the precessing protons generate a tiny oscillating voltage in this coil that gets sent to the preamp and measured.

Polarization Coil Winding

- ▶ Polarization Coil Bottle:
- ▶ This bottle receives a 24-volt DC pulse that drives current through the coil and creates a strong magnetic field. The current that is driven is about 1.5A during the polarization phase
- ▶ This field that is created is a lot stronger than Earth's field which forces the protons in the sensing bottle to align along an axis.
- ▶ After 2 to 3 seconds the current gets switched off abruptly and the protons are left to precess freely in Earth's field.
- ▶ This bottle is created with 21-gauge wire of about 700 turns. 21-gauge wire was used here because it is used to carry a current and the thicker wire won't overheat at the amps required.

Internal Tow-fish Designs

- ▶ Since we focused on getting the actual proton magnetometer pcb to work the tow-fish materials were never purchased but a few designs were made before this decision.
- ▶ The first is an internal sled with 4 holes that would have been connected to the pcb to keep it in place inside the tow-fish.
- ▶ The second design is the outer ring that would of went on around the tow-fish to holder the fins implement some aerodynamics.





Design Reasoning

Overall Reasoning

We chose this two coil – ten module design primarily due to a research paper by the Institute of Physics in Sri Lanka. The paper detailed the creation of a proton magnetometer in detail – we adapted this design to achieve a better sensitivity than what was discussed in the paper, but the general design remained the same

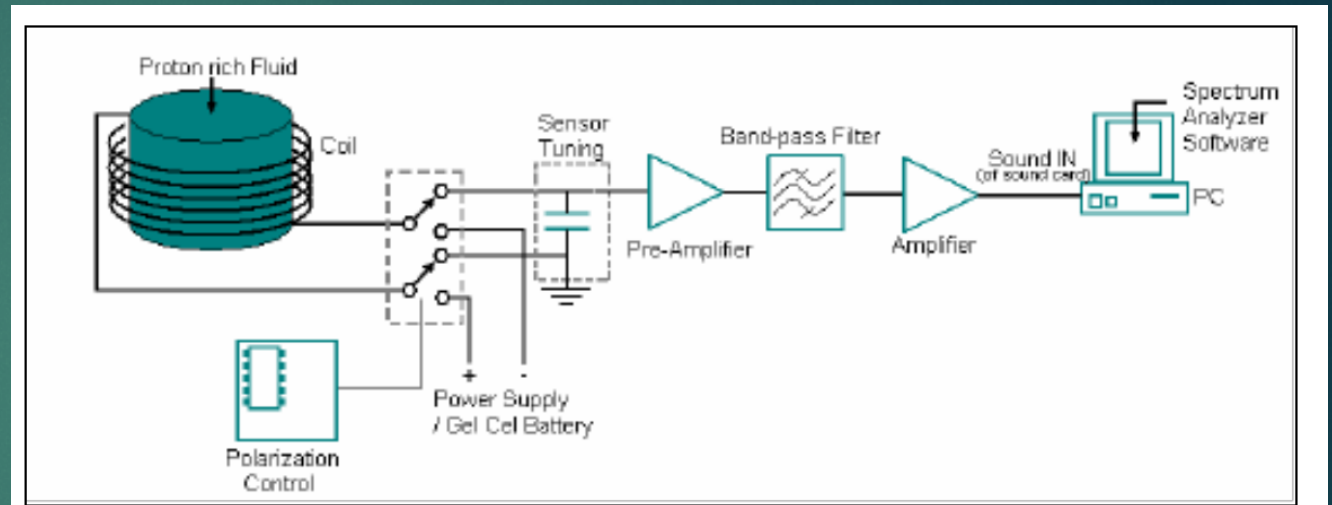
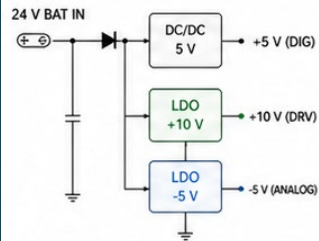


Image Source: [Institute of Physics – Sri Lanka](#)

Specifics Reasoning

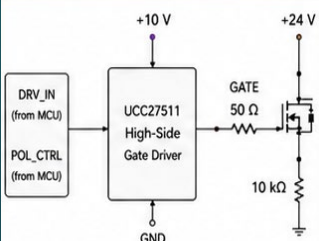
1. POLARIZATION POWER SUPPLY



- 24 V battery input with reverse polarity protection and bulk filtering.
- DC/DC buck converter generates +5 V digital rail.
- LDO regulators generate +10 V (for coil drive) and -5 V analog rail.
- All rails are low-noise and decoupled for clean analog performance.

- The system uses a high voltage source (24V) to ensure the polarization coil can generate a significant magnetic field ($k_{pol} = 75$) despite its 20Ω resistance.
- By using the LM2596 for digital logic and LM317 for analog rails, the design isolates switching noise from the sensitive .5V target preamp signal.

2. MOSFET DRIVER



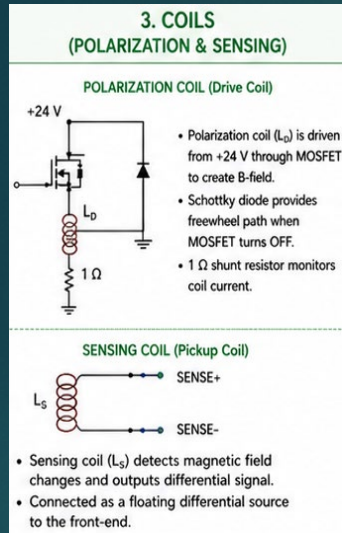
- UCC27511 high-side gate driver switches the polarization MOSFET.
- +10 V supply provides strong gate drive.
- 50 Ω gate resistor controls switching speed.
- 10 k Ω pull-down keeps MOSFET off when inactive.
- DRV_IN = PWM drive, POL_CTRL = enable.

- The UCC27511 was chosen to provide a high-peak current drive to the MOSFET gate.
- To observe precession, the polarization field must be stopped within microseconds ($t_{switch_target_s} = 2\mu s$) to prevent following, which would nullify the signal.

Main Sources:

[Institute of Physics – Sri Lanka ICICEE Conference](#)

Specifics Reasoning

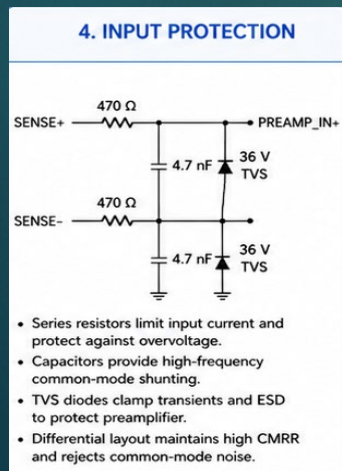


Polarization:

- Using 21 AWG wire for 700 turns allows for a high current density without excessive resistive heating during the .8s polarization pulse.
- The .762m diameter is optimized to create a uniform magnetic flux.

Sensing:

- The higher turn count (2000 turns) and thinner wire (30 AWG) are used here because the induced precession signal is extremely weak - more turns increase the induced EMF according to Faraday's Law.
- The .06H inductance is designed to work in tandem with the input tuning capacitors to resonate near the expected Larmor frequency.



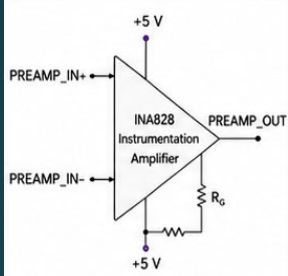
- When the 24V polarization is cut, the coil's 2mH inductance generates a massive back-EMF spike.
- The diodes clamp this spike to protect the sensitive op amps from failure.

Main Sources:

[IOP Conference](#)
[Journal of Instrumentation](#)

Specifics Reasoning

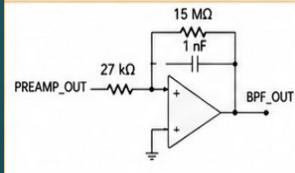
5. PREAMP (INSTRUMENTATION AMPLIFIER)



- INA828 instrumentation amplifier provides low-noise differential gain.
- High input impedance minimizes loading on the sensing coil.
- Excellent CMRR (>100 dB) rejects common-mode noise.
- Gain set by external resistor R_G .

- The INA828 instrumentation amplifier is utilized because the Towfish cable acts as a long antenna for common-mode noise.
- It extracts the tiny differential precession signal while rejecting the massive noise signals common to both input leads.

6. BANDPASS FILTER



$f_L = 2.1 \text{ kHz} \quad (Q = 5)$

- Active band-pass filter centered near 2.1 kHz (Larmor frequency).
- High-Q improves selectivity and SNR.
- Removes low-frequency drift and high-frequency noise.
- Sets measurement bandwidth for optimal performance.

- The ADA4898 stage with 1nF and $1.5M\Omega$ components creates a narrow window around the Larmor frequency ($\sim 2\text{kHz}$).
- This filters out environmental noise (like 60Hz) that the ADC would otherwise pickup on.

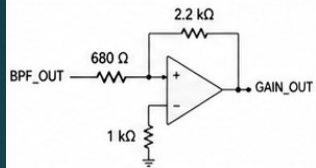
Main Sources:

[Academic Commons](#)

[IEEE GEOSCIENCE + REMOTE SENSING](#)

Specifics Reasoning

7. GAIN STAGE

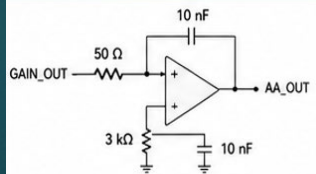


$$\text{Gain} = 1 + (2.2 \text{ k} / (680)) = 4.2\times$$

- Non-inverting amplifier provides additional adjustable gain.
- Sets overall signal level to match ADC full-scale range.
- Feedback capacitor (6.8 pF on PCB) ensures stability.
- Low-noise op-amp minimizes added noise.

- The signal is amplified from the preamp output level (.5V) to a level more suitable for the ADC full-scale range 1.6V).
- Using specific resistors (2.2k and 1k) allows for stable gain that won't change with the temperature changes of shallow coastal waters.

8. ANTI-ALIAS FILTER (LOW-PASS)



$$f_c = 5 \text{ kHz} \quad (2\text{nd Order LPF})$$

- 2nd order low-pass filter prevents aliasing before ADC.
- Cutoff ~5 kHz matches sampling rate (25 kSPS) with margin.
- Protects ADC from out-of-band noise.
- Low distortion, flat passband.

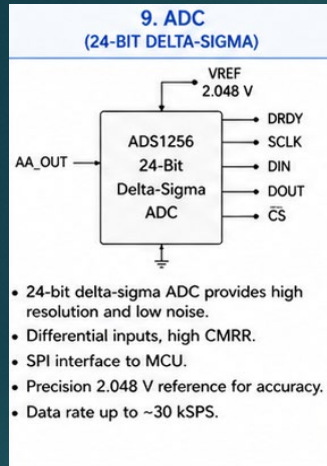
- The $3\text{k}\Omega$ resistors and 10nF capacitors form a low-pass filter to remove frequencies above the Nyquist limit (12.5kHz for a 25kHz sample rate).
- This prevents high-frequency noise from coming back into the measurement frequency range during digitization.

Main Sources:

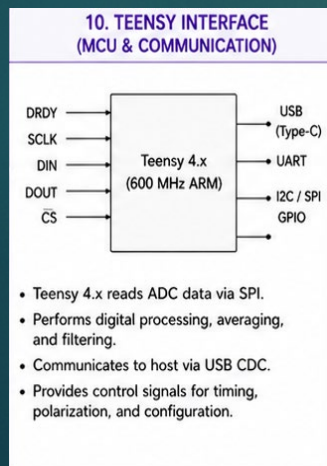
[IOP Conference](#)

[Journal of Instrumentation](#)

Specifics Reasoning



- The AD4003 provides 24-bit resolution, which is necessary to detect the exponential decay of the signal (.5s) with high enough granularity to calculate an accurate frequency.
- The ADR4525 reference ensures that every bit of the resolution is meaningful and not lost to voltage drift.



- The Teensy 4.1s 600MHz clock speed is required to handle the high-speed SPI data from the ADC while simultaneously performing real-time FFTs to identify wreckage anomalies.
- Resistors like R29-R31 33 Ω act as series terminators to prevent signal reflections on the high-speed digital lines between the MCU and the ADC.



Firmware Design

Firmware Architecture Overview

- The TowFish firmware is organized as a modular Teensy 4.1 project. The main sketch, `TowFish.ino`, controls the overall measurement cycle, while dedicated modules manage hardware access, signal estimation, GPS parsing, and SD card logging. `Config.h` serves as the shared configuration layer for pins, timing, and system parameters. This structure improves code readability, maintainability, and subsystem-level debugging.

```
C Config.h
G+ DataLogger.cpp
C DataLogger.h
G+ FieldEstimator.cpp
C FieldEstimator.h
G+ GPSManager.cpp
C GPSManager.h
G+ HardwareController.cpp
C HardwareController.h
G+ TowFish.ino
```

Firmware Architecture Design

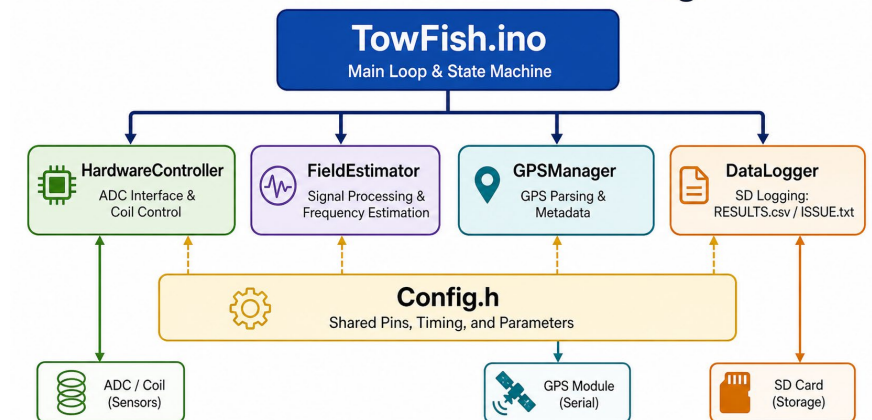


Figure: TowFish firmware organization and high-level architecture overview. Author: Junwen Gu.

Firmware Design Challenges

- ▶ Designing the firmware required more than simply reading sensors and saving data.
- ▶ The system had to coordinate coil polarization, transient blanking, ADC sampling, GPS parsing, signal estimation, and SD card logging within one repeated measurement cycle.
- ▶ Key firmware challenges included timing control, hardware abstraction, synchronized ADC sampling, noisy signal estimation, metadata integration, and reliable output logging.
- ▶ The following slides show how each challenge was addressed through modular firmware design.
- ▶ These challenges were handled by separating the firmware into dedicated modules for configuration, control, acquisition, estimation, GPS parsing, and logging.

Centralized Configuration

- ▶ Config.h stores shared hardware pins, timing constants, and physical parameters.
- ▶ ADC, coil driver, GPS UART, and SD card settings are defined in one location.
- ▶ Measurement timing is controlled by configurable polarization, blanking, and sampling windows.
- ▶ This design makes firmware updates easier when hardware pin mapping or timing requirements change.

[Code: Config.h](#)

```
static constexpr float SAMPLE_RATE_HZ = 25000.0f;

/** @brief Expected Larmor frequency near Earth's magnetic field. */
static constexpr float TARGET_F0_HZ = 2128.8f;

/** @brief Proton gyromagnetic ratio in Hz/uT. */
static constexpr float PROTON_GYRO_HZ_PER_UT = 42.576f;

/** @brief Coil polarization duration in milliseconds. */
static constexpr uint32_t POL_TIME_MS = 2000;

/** @brief Blanking duration after polarization in milliseconds. */
static constexpr uint32_t BLANK_TIME_MS = 20;

/** @brief ADC measurement duration in milliseconds. */
static constexpr uint32_t MEASURE_TIME_MS = 2000;

#define PIN_ADC_CNV      0    ///< ADC conversion trigger / CNV pin
#define PIN_ADC_SCK      13   ///< SPI serial clock
#define PIN_ADC_MISO     12   ///< SPI MISO (ADC SDO)
#define PIN_ADC_MOSI     11   ///< SPI MOSI (ADC SDI, if used)
#define PIN_ADC_CS       10   ///< Reserved chip select pin unless actively
/** @} */

/**
 * @name Coil Driver Control
 * @brief Pin definition for the polarization driver.
 * @{
 */
#define PIN_COIL_POL      7    ///< Coil polarization control pin
/** @} */

/**
 * @name GPS Module Interface
 * @brief Single-UART GPS connection for both read and write operations.
 * @details The GPS module is connected to Teensy Serial3.
 * Serial3 RX receives NMEA output from the GPS module.
 * Serial3 TX sends configuration commands back to the GPS module.
 * @{
 */
#define GPS_SERIAL        Serial3  ///< Unified UART interface for GPS I/O
#define PIN_GPS_RX        15       ///< Teensy RX3 pin
#define PIN_GPS_TX        14       ///< Teensy TX3 pin
#define GPS_BAUD          9600     ///< Default GT-U7 / NMEA baud rate
```

Figure: Centralized firmware configuration for pin assignments, sampling parameters, and measurement timing in Config.h. Author: Junwen Gu.

Measurement State Machine

- ▶ The firmware uses a loop-based state machine to run autonomous measurement cycles.
- ▶ Each cycle separates coil polarization, blanking delay, ADC sampling, and result logging.
- ▶ GPS parsing is serviced in the main loop while state transitions are controlled by elapsed-time checks.
- ▶ During MEASURING, each ADC sample is immediately passed to the field estimator.

[Code: TowFish.ino](#)

```
/**
 * @enum SystemState
 * @brief Top-level state machine for autonomous
 */
enum SystemState {
    STATE_BOOT,           ///< Initial startup state
    STATE_POLARIZING,      ///< Coil is energized to po
    STATE_BLANKING,        ///< Waiting for switching t
    STATE_MEASURING        ///< ADC samples are collect
};
```

```
case STATE_BLANKING:
    if (nowMs - stateStartTimeMs >= BLANK_TIME_MS) {
        dsp.reset();
        stateStartTimeMs = nowMs;
        measureStartUs = nowUs;
        lastSampleUs = nowUs;
        currentState = STATE_MEASURING;

        if (logger.isReady()) {
            logger.logIssue("INFO", "Measurement phase started.");
        }
    }
    break;

case STATE_MEASURING:
    if ((nowUs - measureStartUs) < (MEASURE_TIME_MS * 1000UL)) {
        if (nowUs - lastSampleUs >= SAMPLE_INTERVAL_US) {
            uint32_t sample = hw.readADC();
            dsp.addData(sample);
            lastSampleUs = nowUs;
        }
        else {
            const float frequency = dsp.calculateFrequency();
            const float strength = dsp.calculateStrength();
            const GPSData& gpsData = gps.getData();

            if (logger.isReady()) {
                logger.logResult(frequency, strength, gpsData);

                if (strength < MIN_SIGNAL_STRENGTH) {
                    logger.logIssue("WARN", "Signal strength is low. Check coil or environme");
                }
            }
        }
    }
    break;
```

Figure: Firmware state definitions and loop-based measurement logic for one autonomous measurement cycle in TowFish.ino. Author: Junwen Gu.

Polarization, Blanking, and Measurement Timing

```
switch (currentState) {
  case STATE_POLARIZING:
    if (nowMs - stateStartTimeMs >= POL_TIME_MS) {
      hw.setPolarization(false);
      stateStartTimeMs = nowMs;
      currentState = STATE_BLANKING;

      if (logger.isReady()) {
        logger.logIssue("INFO", "Polarization phase ended.");
      }
    }
    break;

  case STATE_BLANKING:
    if (nowMs - stateStartTimeMs >= BLANK_TIME_MS) {
      dsp.reset();

      if (logger.isReady()) {
        logger.logIssue("INFO", "Measurement phase started.");
      }

      stateStartTimeMs = millis();
      measureStartUs = micros();
      lastSampleUs = measureStartUs;
      currentState = STATE_MEASURING;
    }
    break;

  case STATE_MEASURING: {
    const unsigned long measureNowUs = micros();

    if ((measureNowUs - measureStartUs) < (MEASURE_TIME_MS * 1000UL)) {
      if (measureNowUs - lastSampleUs >= SAMPLE_INTERVAL_US) {
        uint32_t sample = hw.readADC();
        dsp.addData(sample);
        lastSampleUs += SAMPLE_INTERVAL_US;

        if (measureNowUs - lastSampleUs > SAMPLE_INTERVAL_US * 10UL) {
          lastSampleUs = measureNowUs;
        }
      }
    }
  }
}
```

Figure: Firmware timing control for polarization, blanking, and ADC measurement during one autonomous TowFish acquisition cycle in TowFish.ino. Primary author: Junwen Gu.

- ▶ The firmware uses a loop-based state machine to advance through POLARIZING, BLANKING, and MEASURING phases.
- ▶ After POL_TIME_MS, polarization is disabled and the system transitions into a short blanking window.
- ▶ After BLANK_TIME_MS, the estimator is reset and the measurement window begins.
- ▶ During STATE_MEASURING, ADC samples are acquired at SAMPLE_INTERVAL_US and immediately passed to the signal estimator.

[Code: TowFish.ino](#)

Hardware Control Layer

- ▶ Hardware Controller provides the firmware-to-hardware interface for the Teensy 4.1.
- ▶ GPIO pins are initialized for coil polarization control and ADC conversion triggering.
- ▶ The polarization driver is controlled through a single function call: `setPolarization()`.
- ▶ ADC acquisition is started by pulsing the CNV pin, then reading the conversion through SPI.

[Code: HardwareController.cpp](#)

```
bool HardwareController::begin() {
    // Configure coil driver control pin.
    pinMode(PIN_COIL_POL, OUTPUT);
    digitalWrite(PIN_COIL_POL, POL_OFF);

    // Configure ADC conversion trigger pin.
    pinMode(PIN_ADC_CNV, OUTPUT);
    digitalWrite(PIN_ADC_CNV, LOW);

    // Initialize SPI on the dedicated Teensy hardware pins.
    SPI.begin();
    SPI.setSCK(PIN_ADC_SCK);
    SPI.setMOSI(PIN_ADC_MOSI);
    SPI.setMISO(PIN_ADC_MISO);

    return true;
}

void HardwareController::setPolarization(bool active) {
    digitalWrite(PIN_COIL_POL, active ? POL_ON : POL_OFF);
}

/**
 * @brief Trigger one ADC conversion and return the 18-bit result.
 * @return uint32_t Raw 18-bit ADC code.
 */
uint32_t HardwareController::readADC() {
    uint32_t raw = 0;

    SPI.beginTransaction(_spiSettings);

    // Trigger conversion.
    digitalWriteFast(PIN_ADC_CNV, HIGH);
    delayMicroseconds(1);
    digitalWriteFast(PIN_ADC_CNV, LOW);
}
```

Figure: HardwareController initialization and low-level control routines for coil polarization, ADC conversion triggering, and SPI-based ADC communication in HardwareController.cpp. Primary author: Junwen Gu.

ADC Sampling Data Path

- ▶ The measurement loop samples the ADC at the configured SAMPLE_INTERVAL_US.
- ▶ Each sample is read through hw.readADC() and immediately streamed into FieldEstimator.
- ▶ HardwareController.cpp reconstructs the ADC output from three SPI byte transfers.
- ▶ The 24-bit SPI frame is shifted to extract the AD4003 18-bit conversion result.

Code: [TowFish.ino](#) |
[HardwareController.cpp](#)

```
if (measureNowUs - lastSampleUs >= SAMPLE_INTERVAL_US) {  
    uint32_t sample = hw.readADC();  
    dsp.addData(sample);  
    lastSampleUs += SAMPLE_INTERVAL_US;  
  
    if (measureNowUs - lastSampleUs > SAMPLE_INTERVAL_US * 10UL) {  
        lastSampleUs = measureNowUs;  
    }  
}
```

```
// Read three bytes and reconstruct the 24-bit frame.  
raw = static_cast<uint32_t>(SPI.transfer(0));  
raw = (raw << 8) | static_cast<uint32_t>(SPI.transfer(0));  
raw = (raw << 8) | static_cast<uint32_t>(SPI.transfer(0));  
  
SPI.endTransaction();  
  
// AD4003 18-bit data is typically left-aligned in the 24-bit transfer.  
return (raw >> 6);
```

Figure: ADC sampling data path from periodic sample scheduling in TowFish.ino to SPI frame reconstruction and 18-bit result extraction in HardwareController.cpp. Primary author: Junwen Gu.

Signal Estimation Algorithm

- ▶ Incoming ADC samples are first processed with a simple DC blocker to remove slow baseline drift.
- ▶ The firmware tracks peak-to-peak signal strength during the measurement window.
- ▶ Positive-going zero crossings are detected when the filtered signal moves from negative to nonnegative.
- ▶ Linear interpolation estimates the zero-crossing index with sub-sample precision.
- ▶ Frequency is calculated from the number of detected cycles divided by the elapsed time between the first and last crossing.

[Code: FieldEstimator.cpp](#)

```
// 1. DC Blocker (Simple High-Pass)
// Formula: y[n] = x[n] - x[n-1] + 0.995 * y[n-1]
float dcBlocked = 0.0f;
if (_hasSample) {
    dcBlocked = currentRaw - _prevRaw + (0.995f * _filteredDC);
    _filteredDC = dcBlocked;
} else {
    _hasSample = true;
    _filteredDC = 0.0f;
}
_prevRaw = currentRaw;

// 2. Use the DC-blocked signal directly.
float filtered = dcBlocked;

// 3. Track Strength (Peak-to-Peak)
if (filtered > _peakMax) _peakMax = filtered;
if (filtered < _peakMin) _peakMin = filtered;

// 4. Positive-going zero-crossing detection with linear interpolation
if (_totalSamples > 0) {
    // Look for positive-going zero crossing
    if (_prevFiltered < 0 && filtered >= 0) {
        // Linear interpolation for sub-sample precision
        // fraction = -prev / (current - prev)
        double fraction = -_prevFiltered / (filtered - _prevFiltered);
        double exactIdx = (double)_totalSamples + fraction;

        if (_crossCount == 0) _firstCrossIdx = exactIdx;
        _lastCrossIdx = exactIdx;
        _crossCount++;
    }
}

float FieldEstimator::calculateFrequency() {
    if (_crossCount < 2) return 0.0f;
    if (_firstCrossIdx < 0.0 || _lastCrossIdx < 0.0) return 0.0f;
    if (_lastCrossIdx <= _firstCrossIdx) return 0.0f;
    if (_lastCrossIdx > (double)_totalSamples) return 0.0f;

    // Frequency = (Number of Cycles) / (Total Time in Seconds)
    // Cycles = Count - 1
    // Time = (lastIdx - firstIdx) / SAMPLE_RATE
    double numCycles = (double)(_crossCount - 1);
    double timeElapsed = (_lastCrossIdx - _firstCrossIdx) / (double)SAMPLE_RATE;
    if (timeElapsed <= 0.0) return 0.0f;

    return (float)(numCycles / timeElapsed);
}
```

Figure: Signal estimation logic in FieldEstimator.cpp, including DC blocking, peak-to-peak strength tracking, positive-going zero-crossing detection, and frequency calculation from detected crossing intervals. Primary author: Junwen Gu.

GPS Metadata Parsing

```
if (fields[1] && fields[1][0] != '\0') {
    strncpy(_data.utcTime, fields[1], sizeof(_data.utcTime) - 1);
    _data.utcTime[sizeof(_data.utcTime) - 1] = '\0';
}

const bool hasValidFix = (fields[6] != nullptr && atoi(fields[6]) > 0);
_data.hasFix = hasValidFix;
_data.satellites = (fields[7] != nullptr) ? static_cast<uint8_t>(atoi(fields[7])) : 0;

if (hasValidFix) {
    _data.latitude = convertNMEAToDecimal(fields[2], fields[3]);
    _data.longitude = convertNMEAToDecimal(fields[4], fields[5]);
    _data.altitudeMeters = (fields[9] != nullptr) ? static_cast<float>(atof(fields[9]))
}
}
```

Figure: GGA sentence metadata extraction in GPSManager.cpp for UTC time, GPS fix status, satellite count, latitude, longitude, and altitude. Primary author: Junwen Gu.

- ▶ GPSManager extracts UTC time from the GGA sentence for timestamp metadata.
- ▶ The firmware checks the GPS fix-quality field before accepting position data.
- ▶ Satellite count, latitude, longitude, and altitude are stored in the GPSTData structure.
- ▶ Latitude and longitude are converted from NMEA format into decimal degrees for logging and mapping.

[Code: GPSManager.cpp](#)

SD Card Logging Design

- ▶ DataLogger creates RESULTS.csv with structured columns for each processed measurement output.
- ▶ Each completed measurement cycle appends one CSV row instead of storing the full raw ADC waveform.
- ▶ The logged row combines firmware timestamp, estimated frequency, signal strength, and GPS metadata.
- ▶ Latitude, longitude, altitude, satellite count, and UTC time are saved with the measurement result for later mapping and analysis.

[Code: DataLogger.cpp](#)

```
if (!SD.exists("RESULTS.csv")) {  
    File dataFile = SD.open("RESULTS.csv", FILE_WRITE_APPEND);  
    if (dataFile) {  
        dataFile.println(  
            "System_Timestamp_ms,Frequency_Hz,Signal_Strength,GPS_Fix,Lat,Long,Alt,SatCount,UTC_Time" );  
    };  
    dataFile.close();  
}
```

```
dataFile.print(millis());  
dataFile.print(',');  
dataFile.print(frequency, 4);  
dataFile.print(',');  
dataFile.print(strength, 2);  
dataFile.print(',');  
dataFile.print(gps.hasFix ? 1 : 0);  
dataFile.print(',');  
dataFile.print(gps.latitude, 6);  
dataFile.print(',');  
dataFile.print(gps.longitude, 6);  
dataFile.print(',');  
dataFile.print(gps.altitudeMeters, 2);  
dataFile.print(',');  
dataFile.print(gps.satellites);  
dataFile.print(',');  
dataFile.println(gps.utcTime);  
  
dataFile.close();
```

Figure: DataLogger.cpp CSV header creation and result-row logging for storing processed measurement outputs with GPS metadata in RESULTS.csv. Primary author: Junwen Gu.

Firmware Design Summary

- ▶ Firmware is organized into dedicated modules for configuration, timing control, hardware access, signal estimation, GPS parsing, and SD logging.
- ▶ TowFish.ino coordinates the autonomous measurement cycle, while HardwareController manages coil control and ADC access.
- ▶ FieldEstimator converts sampled ADC data into frequency and signal-strength outputs, and GPSManager provides position and time metadata.
- ▶ DataLogger writes the final processed measurement results and associated metadata to SD card output files for later analysis.

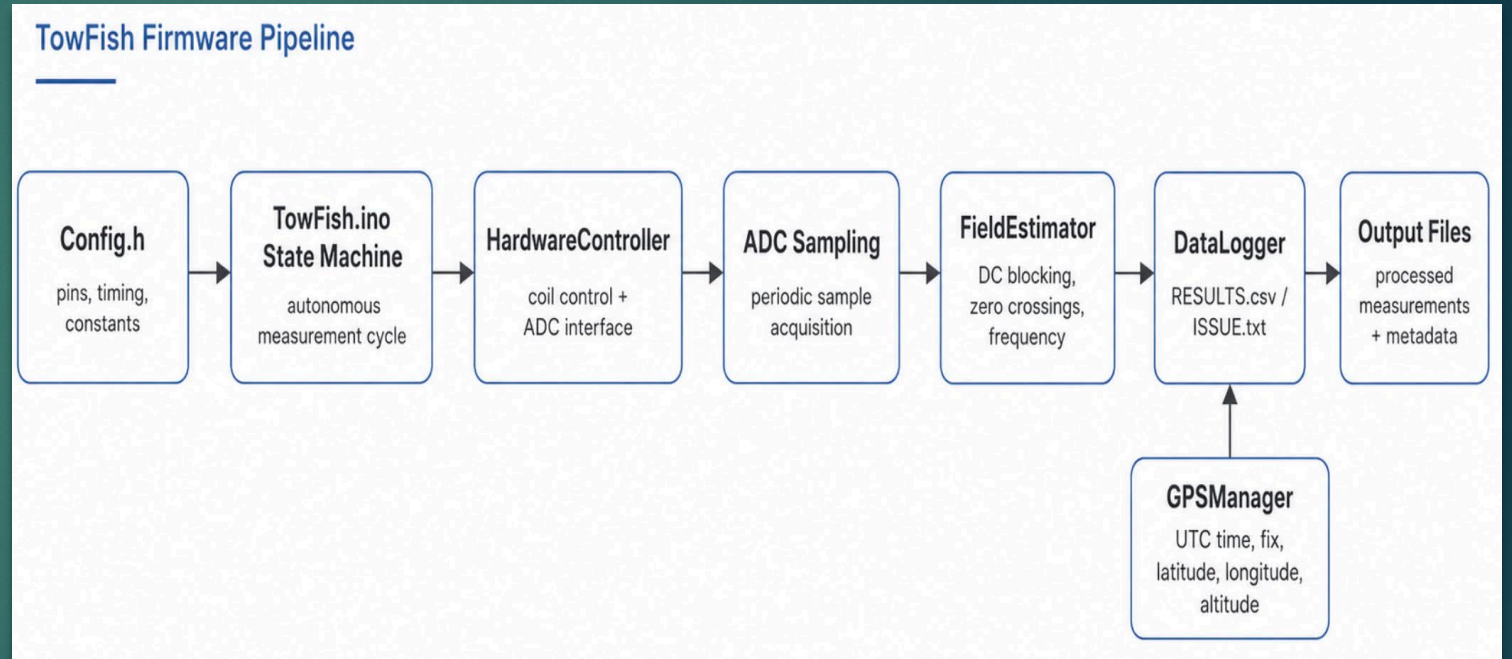


Figure: TowFish firmware pipeline from centralized configuration and state-machine control to hardware interfacing, ADC sampling, signal estimation, GPS metadata integration, and SD card logging. Primary author: Junwen Gu.



Design Verification

Design Outcome

- Ultimately, the design was not entirely functional
- What Worked
 - The polarization coil outputted a directional magnetic field when 24V was applied
 - The sensing coil outputted measurable readings when a nearby magnet was moved
 - The amplification and filtering circuitry worked as intended when a small voltage was applied at the sensing coil inputs
- What Failed
 - When the 24V battery was plugged in, the power supply regulator – specifically the -5V Analog bulk regulator – failed to output the desired value.
 - One of the pins on the teensy was erroneously shorted to ground – leading to massive current draw
 - PCB routing error on both LM2596S bulk regulators – causing them to remain in the off state
 - While some of these problems were easy to solve, due to the interconnected nature of the PCB, it took a while to pin down and fix these problems, and we never ended up fixing all the issues by the end of the semester

Suspected Root Causes Identified During PCB Bring-Up

- ▶ Power rail instability
+5V / -5V analog rails failed to regulate correctly, causing the board to clamp near 3.2 V during current-limited testing.
- ▶ Capacitor polarity errors on -5V_ANALOG rail
Several polarized capacitors may have been reversed; for -5V_ANALOG decoupling, the positive terminal must connect to AGND and the negative terminal to -5V_ANALOG.
- ▶ Incorrect capacitor substitution
Some 100 nF non-polarized ceramic bypass capacitors were replaced with polarized capacitors, increasing the risk of leakage, heating, and rail failure.
- ▶ ADC_CNV / RX1 short to ground
RX1 / pin 0 measured approximately 1 Ω to ground, making the original ADC conversion trigger path unsafe to drive from the Teensy.
- ▶ Possible secondary IC damage
The AD4003 ADC, voltage reference, or analog front-end ICs may have been damaged or overloaded after exposure to unstable analog rails.

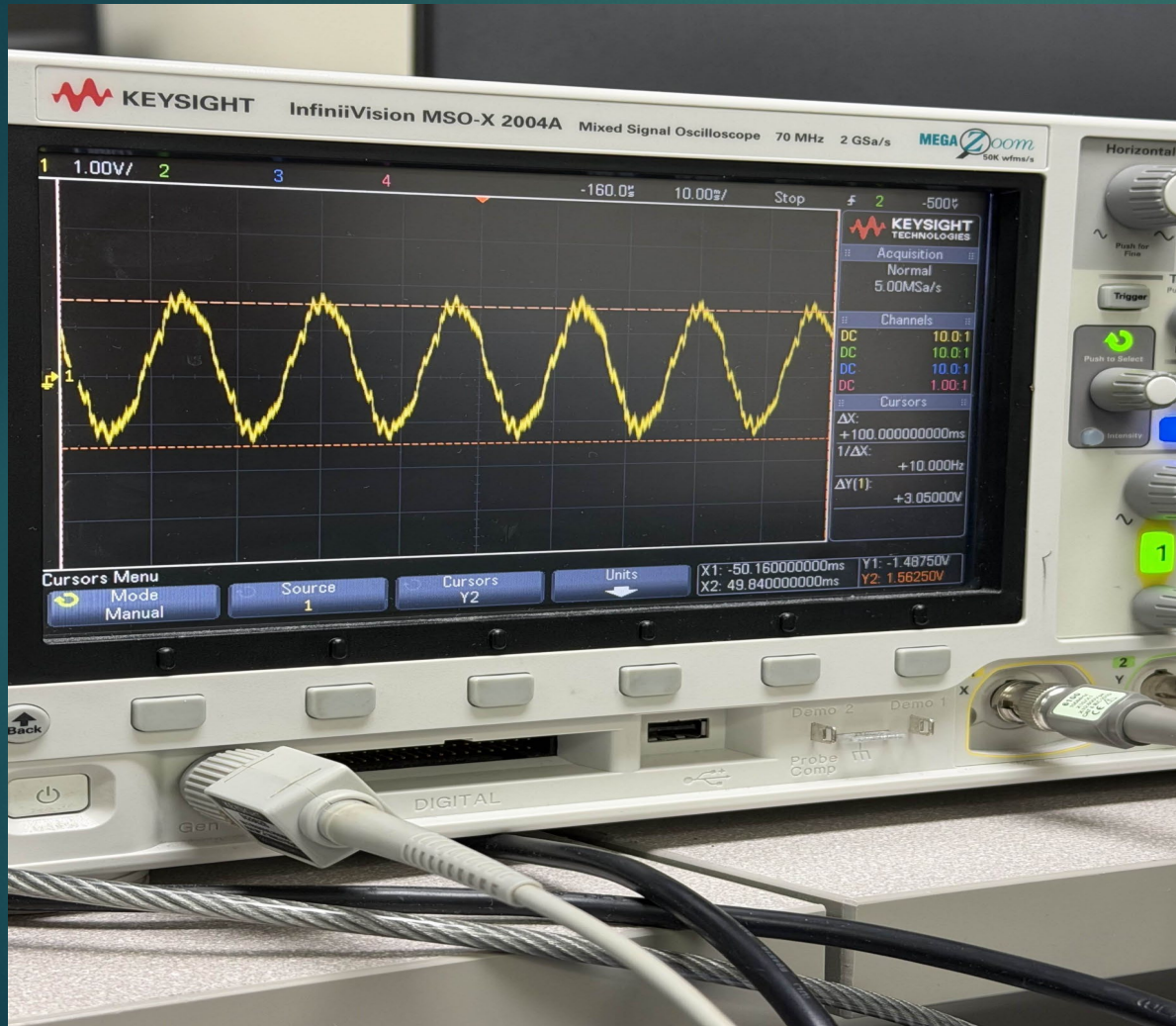
Supported Failure Hypotheses

- ▶ Power rail clamping may indicate abnormal capacitor or rail loading
Capacitor reliability studies show that degraded or failed capacitors can increase leakage, lose filtering capability, and disturb power electronics rail stability. This supports the hypothesis that the 3.2 V clamp was caused by abnormal loading on the analog supply rails. [Teverovsky, NASA/GSFC, 2002/2011](#)
- ▶ Reverse-biased capacitors can cause leakage and destructive failure
NASA capacitor studies state that polarized tantalum capacitors are intended for forward-bias operation only, and reverse voltage can produce high leakage current with potentially destructive results. This supports the concern about reversed capacitors on the -5V_ANALOG rail. [Teverovsky, NASA/GSFC, 2002/2011](#)
- ▶ Incorrect bypass capacitor substitution can weaken local decoupling
ADC and mixed-signal design notes recommend using both bulk capacitance and small 0.1 μ F capacitors near ADC / MCU power pins to filter low- and high-frequency noise. Replacing 100 nF non-polarized ceramic bypass capacitors with polarized parts could therefore reduce high-frequency decoupling performance and increase rail instability. [Texas Instruments, Decoupling Capacitors.](#)
- ▶ ADC_CNV short may create GPIO overstress or latch-up risk
CMOS latch-up literature explains that current injection or overvoltage at signal pins can trigger a low-impedance path between supply and ground. This supports treating the RX1 / ADC_CNV short-to-ground as a serious digital I/O overstress risk. [Texas Instruments, Latch-Up White Paper.](#)
- ▶ Unstable supplies and VREF can invalidate high-resolution ADC measurements
ADC reference literature shows that ADC output code depends directly on VREF, and reference noise or supply ripple can degrade conversion accuracy. This supports treating the AD4003, voltage reference, and analog front-end ICs as possible secondary failure points after unstable rail exposure.
- ▶ [Texas Instruments, How the Voltage Reference Affects Performance, Part 1.](#)

Testing Methodology

- Before ordering the PCB and specific PCB parts, modules 5-8 were tested individually on breadboards with less precise parts.
 - The goal of this was to determine if the amplification and filtering circuitry design worked as intended
 - While the designs did work, some of the schematics were edited due to these tests, as the amplification was not as high as intended
- The power circuitry was not tested before the PCB arrived due to the requirement of specific parts and the long shipping time on those parts – so we opted to test it on the PCB
- After the PCB and parts arrived, we soldered everything onto the board – each component was tested to ensure that nothing was shorted
- Even though everything was interconnected, modules 4-8 were separated enough that these could still be tested separately – the result of that test is shown on the next slide
- The power circuitry testing was done by, at first, inputting a low current 5V signal to ensure regulators were dividing voltage correctly
- After replacing incorrect resistor values and rewiring incorrect traces, we tested with a low current 24V signal to see if the outputs of each component were correct.
- This was repeated until we ran out of time for the semester

Functional Hardware Component



- Output of modules 4-8 circuitry
- 0.3mV sign wave input
- ~1.5V Peak output
- Shows x5k amplification with relatively low noise on output

Firmware Functional Component

- ▶ Simulated proton magnetometer input was processed successfully
The firmware used a synthetic Larmor-frequency signal to verify the measurement and estimation pipeline without depending on the unfinished analog front-end.
- ▶ Real GPS signal was successfully collected
The system captured real GPS metadata, including timestamp, latitude, longitude, altitude, and satellite count.
- ▶ Measurement data was written to the SD card
Each survey entry was saved as a CSV row containing GPS data, estimated frequency, and calculated magnetic-field strength.
- ▶ Logged data was readable on a computer
The SD card output could be opened and inspected on a computer, confirming that the logging format was usable for post-processing and documentation.
- ▶ Firmware workflow was functionally verified
The test confirmed the core firmware path: simulated signal input → frequency estimation → GPS metadata collection → SD card logging → computer-side data review.

--- NEW SURVEY SESSION ---							
Date	Time	Latitude	Longitude	Altitude_m	Satellites	Frequency_Hz	Field_nT
NO_DATE	5793	0.0	0.0	0.0	0	2128.7983	49999.9609
2026-3-25	21:6:46	43.07190	-89.409223	418.30	5	2128.7993	49999.9844
2026-3-25	21:6:50	43.07190	-89.409240	421.10	5	2128.7979	49999.9492
2026-3-25	21:6:54	43.07190	-89.409266	424.00	5	2128.7993	49999.9844
2026-3-25	21:6:59	43.07192	-89.409273	422.00	5	2128.7998	49999.9961
2026-3-25	21:7:3	43.07178	-89.409084	415.20	5	2128.7993	49999.9844
2026-3-25	21:7:7	43.07174	-89.409049	414.00	6	2128.7993	49999.9844
2026-3-25	21:7:11	43.07174	-89.409040	414.30	6	2128.7974	49999.9375
2026-3-25	21:7:15	43.07174	-89.409023	411.90	6	2128.8013	50000.0313

Figure: CSV output showing simulated magnetometer measurements combined with real GPS metadata, written to the SD card and verified on a computer.

Author: Junwen Gu.

Suggested Solutions

- Test every subsystem on a breadboard before ordering the PCB – especially the power circuitry.
- Build and test the power regulation circuit separately using the final exact components before PCB fabrication.
- Add test points for all major voltage rails, regulator enable pins, sensing coil outputs, and amplifier outputs.
- Review PCB routing carefully, especially regulator enable pins, ground connections, and Teensy I/O pins.
- Verify all resistor divider values and regulator feedback paths before soldering parts.
- Add current-limiting protection during initial power-up to prevent damage from shorts.
- Use LEDs or simple indicators on key power rails to quickly confirm regulator status.
- Add fuses to reduce damage from accidental shorts.
- Order critical parts earlier so the final power design can be tested before PCB manufacturing.
- Use a modular PCB layout where modules can be physically separated or independently powered.